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PLANET stories

A.N.C.

SUMMER



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by **POUL ANDERSON**

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A BOOKLENGTH NOVEL OF THE INFINITES

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*** START OF THE PROJECT GUTENBERG EBOOK DAWN OF THE
DEMIGODS ***

DAWN OF THE DEMIGODS

By RAYMOND Z. GALLUN

As unheralded as ghosts, but as significant as a new dawn of history, there came to Earth from distant Ganymede's glowing crescent—three micro-androids, minuscule beings, carrying the moot treasure of immortality.

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Somebody invented the first locomotive. Then came the nuclear bomb. I guess that people were somewhat scared of newness both times.

Mostly, it has been worse ever since.

World War III was also before my day. But then fear, the protective emotion, played a reasonable part. So no cities were actually vaporized. But our side came out the victors with bombers so high-flying that they were already atom-propelled rocket ships of space. We had artificial satellites circling the Earth, and a fortress on the Moon.

I missed the first exploration of the solar system, too. There was hot Mercury, carbon dioxide-smothered Venus; Mars and its ruins and quiet colors; and what was left of Planet X, whose people destroyed the Martians in war, though their planet itself got blown all to bits in the same struggle, its fragments now being known as the asteroids.

The moons of Jupiter and Saturn were also invaded by men, as were the frozen-methane-and-ammonia blizzards of Uranus and Neptune, and the frigid mountain peaks of Pluto, farthest world of all.

There were always yarns about "Little Men" and whatnot, of course. Yet no contemporary intelligent races were found across space. There were just queer skeletons and dried up corpses millions of years old. Rusting on Mars, or floating free and broken among the Asteroids, were the remains of inventions, and other cultural evidences. Space ships had wandered as far as Pluto during those past ages, too; and various relics were left on this sphere or that. Scientific study of these things meant more speed for our technical progress in medicine, atomics, metallurgy, almost anything you could mention.

Three cheers for us, and wasn't progress wonderful? But I guess plenty of folks felt dumb and slow and confused.

I, Charles Harver, was born in Chicago, March 9th, 2014. But in my earliest, murky memories, Earth was only a place known from television, picture books, and the nostalgic remarks of my parents. We had a house and a flower and vegetable garden under a transparent airdome of dark blue plastic. The sun would shine among the stars for what I heard was fourteen

days; then, for another two weeks the solar lamps would burn in the dome top.

The region where we lived was called the spaceward lunar hemisphere. Earth never shone there, but life was good. There were other kids, and school, and the usual dreams about being a bold space wanderer, speeding out to find unimagined marvels.

Dad was a technician in the research labs, just a few miles from our house by tube train. I could see the walls of the buildings in the bleak volcanic distance.

Dad used to pretend he was wrestling with me. "Charlie," he'd say, "a kid better grow up tough and flexible these days. Not mean but rugged—ready for anything. Don't ever go soft on me, Charlie, with all the temptations of modern comforts. You know one thing the labs are looking for, *already*? Yeah, a way to reach the planets of the stars! Maybe a means—and an engine powerful enough—really *will* be invented to force a shortened, interdimensional path across the light-years if the structure of space itself doesn't burst under test! Keep your head down, kid! The work is much too dangerous to be conducted on the densely populated Earth. There could be an awful blow."

Dad was a big, dark man. Talking like that, he looked thrilled and scared.

Dad used to bring Dr. Shane Lanvin out to our house to enjoy Mother's cooking, which, even considering the aid of a fine automatic kitchen, was something special. Dr. Lanvin was a wispy little man with a ragged blond mustache. He was much older than Dad; though Mom used to say that even if he was a famous scientist, he was part child in his interests; that he liked even toys.

"I always did enjoy things in miniature, Lillian," he'd admit.

Dr. Lanvin was an instrument specialist, which meant designing and assembling parts that you could scarcely see, so small were they. Once he made me a toy. It was a ball that absorbed the energy of sunshine, and rolled after me wherever I went, in the plastic-sealed, tree-lined streets of

the lab staff housing area. Following the sounds of my footsteps, it seemed half alive. Maybe it was a forerunner of all that was to come.

Once Dr. Lanvin showed me a bit of quartz, like a grain of sand. It was mounted in a little round case, fitted permanently to a powerful pocket microscope. Through the scope you could see one flat face of the quartz grain, glinting. Carved on it, unmistakable, were horizontal rows of symbols.

My spine tingled. "Did you engrave them, Dr. Lanvin?" I demanded. I was eight or nine, then.

"I could have, Buggsie," he answered, using my nickname. "With a micro-manipulator and a diamond-chip. Only I didn't."

"Then who did?" I pursued.

"I wish I knew," he replied. "A friend of mine collected some two thousand of these tiny, non-ferrous—not iron-bearing, that is—meteors, floating free in the asteroid belt, and mailed them to me. My microscope revealed this unusual one. The symbols are about the same as those used by the beings of Planet X in their full-sized inscriptions, before some vast nuclear charge from Mars blew up their world. But no man can read such writing. That's about all I know—yet."

This remained almost our only information on this particular subject, until years later.

We might all have been blown to Kingdom Come there on the Moon, had any of the lab experiments gotten seriously out of control. There were minor blasts. But I lived out my time there safely. I even worked in those labs myself for several months, and by then even the stars seemed technologically nearer. Dr. Lanvin had left the Moon, accepting a professorship at the University of Chicago, and it was soon decided that I'd be sent there to complete my education.

Mom said an odd thing as she and Dad saw me off at the Tycho spaceport: "I wish we were going there, too, Charlie. I wish we had a little country place, far off from everything, and a cow and some chickens."

"That's a primitive mouthful for a modern woman, with no idea of modern farming, let alone such an antique setup," Dad chuckled. "Well, sometimes I

yearn for simplicity, too. We're weak, slow-adjusting characters, left a little behind by the onrush of the times. So long, Charlie, watch yourself. It seems funny that I, an Earthian, have a son who'll actually have to get used to Earth."

Yes, that turned around situation, characteristic of our era, seemed odd. I was nervous about my big, sophisticated, native planet, as if it was an alien world! As a youngster, men had kidded me that I couldn't even endure its huge gravity!

The new university outside Chicago proper was beautiful. As arranged, I went to live at Dr. Lanvin's house. I adjusted to Terra, even though, within twenty-four hours of my arrival, there was a catastrophe that couldn't have happened in a previous age.

A freighter coming from Mars to the Chicago spaceport, couldn't decelerate. Nobody knows completely what errors of human stupidity were committed aboard the doomed ship under the goad of panic; but the *Venetian Prince* came down like a colossal meteor, fortunately in almost open country miles from the port. Yet a town of fifteen thousand nearby was wiped out. At Dr. Lanvin's house we felt the shock wave and the hot wind. The western horizon glowed red. No doubt a crater will stand for ages at the site of the crash.

As I watched from Doc's dooryard, all the loved romance of space and the future seemed to turn sour on me.

"Disasters that afflict innovations always affect people about the same, Charlie," Doc breathed heavily, not calling me Buggsie, now. "Train wrecks, the sinking of an ocean liner, the crash of a great plane. Now a space ship destroying a town. The magnitude just gets bigger, more terrible. There'll be an investigation, terror, grief, complaints; laws will be changed. But wider and better interplanetary travel will go marching on, with everything else."

I got in on the ground floor of Dr. Lanvin's work in advanced robotics. Robot devices had been used for various purposes for many years. But Doc had invented some much improved ones. I tried handling several. Then, like part of my obscure destiny, the chance came to really prove one of them.

Fire broke out in an unrenovated warehouse near the edge of the city one night. Doc and I drove to the scene in his autocar. There was a lot of inflammable and possibly explosive material. Someone shouted that a watchman had been overcome by smoke inside the building.

"Get him out, Charlie," Doc said. "Your body is more agile than mine; your control of an artificial one will be the same."

Sitting in the car, I put the control helmet over my head. In it there was no old fashioned television screen, and no complicated guide levers. What the helmet did was detect and sidetrack the motor impulses from my brain, broadcasting their pulsations by short wave radio to the robot, which I thus guided as if it were my own form. Similarly, sensory impressions were radioed back to the helmet, there to be reconverted into impulses directly perceptible to the sensory centers of my brain, without the intervention of my eyes, ears, and other sense organs.

So, in effect I was living in a shape not handsome in a human way, stronger than my own, and far less limited. Like a demon I stepped out of the rear seat of the autocar on asbestos-shod feet. Propelled by steel muscles energized by a motor drawing current from an atomic battery, I walked past less intricately robotized fire-fighting equipment. Through smoke that would have strangled an unprotected man, I climbed a ladder and went through a window from which a plume of flame belched. I felt no inconvenience whatsoever. There was a thrill in that—like being something super.

After that I was a bit lost. But a voice growled instructions near me—in the car, that is; I had almost forgotten that I was really there, and not in the blazing warehouse. Muffled and harrassed, it reached my own ears through the control helmet:

"Walk your robot inward, kid, for cripes sake! Follow a beam if there's no floor left! There'll be a little office room...."

I knew that it must be some chief of the fire fighters who was giving me directions.

With flames all around, I—or the machine—scrambled along a steel support, and through an opening in an inside wall. Flames had not penetrated there, and automatically I saw through the opaque smoke by radio waves sent out by, and bouncing back to eyes that belonged to the robot; parabolic antennas, they were. The images were visual and unblurred, and lacked only color.

I found the office, and the man who had collapsed there. I pressed an oxygen mask from my insulated pack over his face, and wrapped him in an asbestos blanket that I carried.

"Rush to the main door of the building, kid!" the voice growled again. "I think the wall of flame is less deep there!"

Doing it was a cinch, though I went through a hundred feet of pure fire in two great leaps. I dropped the guy on a stretcher outside the door. Let the medics work on him. I had to remind myself that he had been rescued, not by me, but by a product of science.

Back at the car I made the robot polish the soot off itself with a cloth, and then climb into the rear seat to assume an inert position for transport, again. After that, I removed the helmet.

"Well, Charlie, another foretaste of the future, eh?" Doc said from behind the wheel. "Make way for tomorrow...."

"Yeah," I grunted raggedly. "Like being more than human."

The guy who had been giving me directions still stood beside the car. Somehow, I sensed that our innocent remarks were wrong things to say in his presence. I studied him from my six-feet-three height. Growing up in low lunar gravity, a fellow shoots up amazingly.

His face, topping a massive body, was beefy and rough and kind; but now there was fury in it. He was like a tame bear after being baited and confused for too long. And he'd just been through some nerve wracking minutes of responsibility for a man's life.

"Okay, kid," he rumbled. "You and science saved that man. Thanks. Otherwise, it's all the same. Tomorrow, and more gadgets! Nothing stays put so a guy can understand it! The world just rushes on, till maybe not even gravestones mean anything anymore—except when a spaceship rubs

out a town, killing my brother, his wife, and their four kids, among the others! Pretty soon you think that the whole universe is going to fall apart, with all the junk in it, and that there aren't going to be any real folks anymore! When all you wanted was peace for your family. Then you get all mixed up, and want to kill and smash whoever and whatever makes this so. Damn! Dammit all!..."

I held my hands poised to grab him if he tried to jump me. Only his grief kept me from calling him a fool. Yes, he might have attacked me, except that now he went all to pieces. Big sobs wracked him.

Doc and I didn't have to do anything because two cops came and led him away.

Doc shrugged sadly. "Neurasthenia. It's getting commoner, Charlie," he said. "A straw just broke another camel's back. Our friend has had a tough time. Besides, he's one of the slow ones. Slow to adjust and grow with his civilization. Oh, he'll probably straighten out. Or the cerebral specialists will fix his brain. He'll be an easy going, untroubled individual. Is it right and democratic to tamper with a man's mind? Well, would you let him be insane, poignantly miserable, for keeps?"

Again I had a primitive qualm. The Earth was all around me, strange, teeming, overpopulated in spite of colonies across space. The crowd, jubilant over the robot's demonstration, was all around us. But I bet that every one of those people was at least a little bewildered by the times.

"Let's drive home, Doc," I urged sharply.

II

As we whizzed along, Dr. Lanvin smirked at me like a sly elf. "To what our poor friend complained about, I owe much," he remarked. "Consider my birth-date, January 23rd, 1932. It's now 2033. Yes, I'm a hundred and one, though I look and feel fifty by old standards. It's common enough. Wizardry? No. Let's face facts, Charlie. Something like immortality has been sneaking up on the human race for well over a century. First, diseases

were conquered one by one. Meanwhile, surgery, replacing worn out organs with new ones grown artificially, went far ahead. Hormone therapy was developed. The final degenerative disease, senility, is proving to be just as conquerable as cancer. Remove its causes—accumulation of minerals and certain fatty acids among other things, and tone up the machine—and it just isn't there anymore!"

Doc paused for breath, then went on:

"Yes, there's plenty that we don't yet know about the wonderful mechanism of the human body. But we don't need to know everything to keep it living on and on. Because, with a little help, it restores itself. The trouble with our viewpoint is that death has been the destiny of all life on Earth for so long that it seems like an inviolable tradition. A silly attitude, don't you think? Now, have I disoriented you some more, Charlie? Don't be embarrassed. I feel somewhat that way myself. Maybe your mood is right for me to go a step further into the murky Destiny of Man, eh?"

My hide was tingling with something like dread. But I was eager. "I'm ready for anything, Doc."

We got back to his old house under the trees of the campus. From a cabinet in his quiet living room he took a plastic box. In it was a small, oblong bar of pinkish substance that wriggled slightly, as if it were animated.

"Touch it," Dr. Lanvin commanded.

I obeyed. The stuff was warm, and in response to contact with my finger, it writhed violently. "Unh—what in hell!" I grunted.

"It's something a big commercial laboratory managed to produce for abstract reasons," he answered. "It isn't any one substance, but its structure does include quite a few complex silicone compounds. Chemically it's not static. Processes and structural changes are going on inside it constantly. Its microscopic texture is cellular, like animal tissue. Pour, say, sugar dissolved in water on it, with the addition of certain salts, and it absorbs the solution slowly, along with oxygen from the air, to produce a kind of tissue-combustion, heat and movement. But it can convert sunlight, or simple heat from an outside source, or electricity, into motion, too. And it grows. Cut a piece of it off, and that will grow, too, as if reproduction had occurred. So—

would you call it life of a sort? It's a lot more rugged than common life. Here, I'll show you, Charlie...."

Doc picked up a small soldering tool. When its point glowed red hot he held it close to that pinkish oblong. It did not recoil from the heat. Instead, as if impelled by some inherent automatism or instinct, it curled itself around the tool, and, hissing softly, seemed to enjoy the warmth. When Doc switched off the current, it uncoiled itself as if in disappointment. It wasn't burned.

"Cold it is equally resistant," Doc remarked. "Especially when its vital fluid, moistening it inside and out, is changed to something with a lower freezing temperature than water. Alcohol, for example, or liquefied air or ammonia gas. Then its chemistry, and the flow of energy continue on a different temperature plane, for it is supremely adaptable, Charlie."

Dr. Lanvin's sly expression matched the chill along my back.

"Okay," I growled. "Now tell me what you're *really* thinking."

He shrugged. "Oh, nothing definite, Charlie. Someday reaching the stars in another figurative sense, maybe. As is, this stuff isn't of much use. Call it 'protoplast' as its creators do—a tougher, upstart brother of protoplasm—life. It isn't molded. But what if, in a vastly improved form, it could be someday?"

I frowned. "An animal?" I questioned. "Artificially made? Or—a man? An android, that is? Pure fantasy, of course, yet. A robot, with a robot's ruggedness, but made completely in human form. Servants maybe?"

Doc Lanvin's mild grin turned crooked. "Servants?" he challenged. "Is that all? What if we were living the last century of man's existence as original man? No, I don't necessarily mean the often dreamed-up possibility of a robot conquest of humanity by force! But what of the 'improved model' principle, applied by humans to themselves, with the transfer of mind and ego to a body that could live without harm in the cold vacuum between the planets, or in an inferno; a body unaging, and destructible only by absolute violence? No, Charlie, this development must normally be a long ways ahead. But what if?"

A cold tingling had started around my heart, spreading inexorably to the tips of my fingers, toes, and tongue.

"Doc, I don't know," I said slowly. "To the flexible of viewpoint—wonderful. But it might be the ultimate shock to those who want tomorrow to be understandably and reliably like today and last year. To them it might be a hell; the death of everything reasonable, and a catastrophe to resist with all the weapons in the modern armory, and with the last fury of dying brain and muscle."

"I thought you'd react something like that," Doc sighed.

My laugh was unsteady. "Then why don't scientists stop digging? Nature can bite back."

"We can't stop, Charlie. Like everything, man is part of nature. He was given wits and curiosity to know the whys and wherefores of everything. It's like a religion—trying to learn a little more about, and get a little closer to, Whatever It Is That Keeps The Cosmos Running. Or you can say that all of man's works are works of nature, with him as the tool. That is our oneness with the universe which we've got to grow with. The fears are often childish. I feel the scare, too, Charlie, but I think you're like me."

"I hope I am," I stated.

"Thanks. Go to bed, Charlie. We've gabbed enough for now."

"Nix," I answered. "I think that maybe you have been leading up to some mention of your own work, Doc."

Dr. Lanvin's fingers tightened on the arm of his chair. "All right, Charlie," he said. "Down at dust-grain size is my own segment of the universe—my miniature region to explore—as others mean to explore the planets of the stars. It's a weird zone where familiar physical laws are curiously altered in their effects by relativity. Humans can't go there in their own flesh, at least not yet. But I believe that there may be a simple if difficult way to build a tiny metal proxy, operated the same as you operated that fire robot. Then, perhaps some compelling mysteries will be solved."

"For instance?" I prompted.

Doc nodded toward a photograph on the wall beside the old fashioned picture of his former wife. The first photograph showed his tiny pink meteor. Its much magnified hieroglyphics seemed to wink at us enigmatically.

"How that writing got there," Doc answered. "And now, more. Government authority has asked me to help, Charlie. From Ganymede, largest moon of Jupiter, comes a report of a cache of tools in a chest that itself is of almost microscopic dimensions. Finding it, several men were afflicted with dizziness and fainting. One died. Autopsy revealed many little seared, reddish lines crisscrossing inside the cortex of his brain. Also, in the asteroid belt, several space ships have gone out of control, the finest parts of their most delicate control mechanisms severed as if by intense heat."

"Beings," I breathed awedly. "The implication is clear but crazy. Why, a being no smaller than a rat couldn't have human intelligence. The molecules of matter remain of the same size. Building a truly sentient brain at such an extreme of smallness, using those same molecules, would be like trying to make fine pottery out of coarse sand!"

"Who said anything about molecules?" Doc demanded almost angrily. "When—just for instance—the flow of far smaller electrons has been the soul of complicated calculating devices for a century? But who knows anything? Maybe, before long, I'll let you in on my project, if you have the courage and interest. But you need more general education. Now, not another word. Go to bed. Answering your whys and whats, I'd be mostly guessing, too."

I went to my room where I lay for hours with a tingle in my guts, aware of the frosty stars, the squawking katydids, the universe big and little, and buzzes from Doc's workroom. And I thought of how the human body conformed to the laws of machines. Hence it was a natural robot. It was near dawn before I slept.

Thereafter, life went on in the sunshine and shadow of a campus, idyllic and gentle, masking unrest. There were my instructors and the classrooms and labs, and the faces of my contemporaries, easy going young faces, matching a languid attitude of body coupled with latent strength, as a cat's languidness is coupled with the capacity to galvanize into lightning speed.

For them the times held a pleasant spark of fright, and the rich red meat of coming triumph. In all fairness, I was one of them.

Minden, Fellows, Bowhart, Griswold, Scharber, and the others—the rhythm and meaning of their voices and words was usually the same:

"Hi, Charlie! How was the astrogation quiz?" Or: "Yesterday, I rocketted the training ship two thousand kilometers over Lake Michigan, and it was a cinch!..." Or: "Could any course be dopier than this 'Suggested Techniques For Establishing Friendly Relations With Possible Extra-Terrestrial Intelligences?'" Or: "So long, Charlie! I'm going out to help build Pallas City for the asteroid miners!..."

Or it might be Mars, Venus, Mercury, or the well remembered Moon, where huge, fantastic starcraft were already in the blueprint stage.

Yes, all this was my life, too; though already it seemed half diverted to another, much stranger destiny.

And there was Janice Randall whom I first spotted in Astrogation Lecture.

But I really met her in the company of George who had a special room up in the top of the University Library. It was an eerie place, suitable for such an entity. George was more than an electronic calculator; he was a Giant Brain. He was also a student's and research worker's oracle. You could ask him questions.

One day, with Doc Lanvin's remarks sharply in mind, I went up there and waited my turn at George's microphone.

"Can a true android, with all the human attributes of mind and feeling, be made, George?" I asked.

George rustled inside his unpretentious black cabinet and replied in a bass voice:

"I believe that it can and will be done. Like space travel, it is part of the natural course of history."

"What is the most difficult phase of doing so?" I enquired further. "Building the brain?"

"No. Building the brain should be relatively simple. Giving an android a consciousness, an awareness of self, should be much more difficult. Philosophers have had trouble even defining consciousness."

I chanced a third question, the answer to which I already knew: "Don't you have a consciousness, George?"

"It is understood that no means yet exists to provide Earthly mechanisms with such a thing," George replied evenly. "I am no more aware than the first crude adding machine thinking out the sum of five and seven. A question to me merely sets a search and a reasoning process for a reply in motion. It is not necessary that I *know* that I do this."

Someone spoke from behind me: "Unaware thought. It even happens in our own subconscious minds. But it's hard to believe that George doesn't know his own reality. He's so human."

I turned. It was Jan Randall, coloring a little. "Oh, I'm sorry Mr. Harver. I didn't mean to eavesdrop," she said.

"Eavesdrop?" I chuckled. "There was nothing private about my question. Go ask yours. I'll wait out of earshot."

"I haven't any," she answered with a smile. "I come here often just for the mood. This place feels like a temple; as if God and all nature were here. George isn't much of all that, but he seems the best contact. Now, shall we both laugh?"

"Let's feel awed and humble, instead," I replied.

After a pause, I asked lightly: "George, is it all right for Janice Randall and I to have dinner together?"

George was small. Always, he refused to give out social advice. "This, I am not permitted to answer," he rumbled.

Jan and I laughed gayly together as we turned to leave. Jan was unobtrusive, but very pretty. Her hair was light brown, her features were fine, her nose turned up, her height reached to the center of my chest. And she had her eyes on a spacewoman's career.

From beside the door a pair of slightly fanatical eyes under a high forehead smirked at us. The jaw was strong; the smile was crooked, humorous, gentle.

"Hi, Cope," I greeted. "What brings you here?"

"This I have to watch, Harver," he answered. "The machine telling the man—already. Screwballs! Where will it end?"

"Who was that?" Jan asked as we were going down stairs.

"An English-Lit classmate of mine," I answered. "One who believes that all virtue is the past's. Call him the conscience of the human race. Armand Cope."

Little hard glints showed in Jan's eyes before she said, "Oh," mildly, and laughed.

After dinner I took her to meet Dr. Shane Lanvin. Six months later he said to both of us:

"Like a hiring officer picking a starship crew, I have to look for guts, wits, and certain other qualities, in prospective helpers for what I am attempting. There may be danger. And I wouldn't want anyone to go soft and back out later. So here's your chance—part-time for the present. But I want a final yes or no."

Mild Doc Lanvin could be hard. But he knew Jan's quality. She had taken courses of study and training, of which nine-tenths of the students were masculine. Her reactions to tests for quick thinking, emotional ruggedness, and physical stamina, were all good. Moreover, she had excelled in the study of instrument making, with which we had both occupied much time. Her manual precision was better than mine.

"On your terms, count me in, Dr. Lanvin," she said quietly.

"Even without Jan, I think I'd be a foregone conclusion, Doc," I told him.

So, with every minute that we could spare from our regular studies, we were working with the great specialist of the miniature, trying to push another frontier downward into The Small. Doc had his duties to the University, but he had his nights and weekends, and the additional drive of the odd and grim reports which had already come from deep in space.

Do you know what a micro-manipulator is? It begins with a simple, high powered microscope. But in its field of view are mounted little slide rods, fitted with hand operated vernier screws, by which they can make movements so fine that a gesture of a thousandth of a centimeter may seem the widest of swings. Attached to the slide rods are forceps, and measuring and cutting instruments, some too small to be visible to the unaided eye.

Under one microscope Doc had even set up a real, power driven lathe, a quarter-inch long. Under another was a sort of assembly area. There, a shiny robot, half an inch high, with all the intricate control and circuits combined into it, was taking form. Cables were as fine as spiderwebs.

III

Reproducing that first robot in triplicate was easier than it might seem, for when we had set up all the small machinery to make the parts, duplication was almost automatic. But assembly remained a tedious chore. On the other hand, the control hoods were almost of the standard type used for much larger automatons.

Still, it was eight months before Doc announced on a Saturday afternoon: "Step one completed. Now to repeat in step two!..."

We had test proved all three robots as soon as each was ready. But now that each of us possessed a metal proxy, we could all go as a group on that first step down into The Small.

Sitting in chairs in Doc's workroom we put on our control hoods. Then, sensory illusion seemed to make us leave our real bodies behind. The top of the work table spread suddenly around my tiny, artificial eyes becoming a vast, cluttered plain. The ceiling was our sky. The fluorescent lights were multiple suns. Doc and Jan were shining, man-tall monsters, exactly like myself. I couldn't tell them apart, until manner of speech betrayed Jan.

"Look at us!" she shouted gleefully in a thin, buzzing voice from a tympanum in her chest. "Coming this far is like dropping into an abyss, half

way to the bottom! And see the *real* us! Great, hooded colossi, sitting as if asleep, in the distance!"

"Yeah, I know, Honey," was all I said.

In this moment of half realizing a goal, Doc's love of miniature things became tense impatience.

"All right, my worthies!" he buzzed. "Supper and being ourselves again is only a few hours away. So let's get started on tougher step two!"

We hurried to a clear plastic box (of building-size to us now) and inside its drilled doorway our materials were waiting.

There were the roughed out beginnings of other micro-manipulators, except that now, for work on pieces smaller than the width of a single light wave, the microscopes would have to be of the electron variety. Their parts had to be polished and fitted together here; for even that was labor beyond the direct doing of a man in his own flesh. Now we had to finish a whole array of super-fine micro tools and equipment—lathes, heaters, shapers of glass.

Not until then could the real work proceed—making robots of which only the largest pieces yet existed, still in the rough. They would be robots bearing the same size-relationship to our present half-inch selves, as those same selves bore to human beings!

"Specks, dust particles, dimensions on the order of the smallest insects," Doc's new self buzzed. "Down near the barrier, the limit of smallness, beyond which metals become, by relativity, too hard, and too coarse-grained to be shaped. Small objects are always relatively stronger than large ones. Yes, you'd have to decrease the strength of materials in proportion to size to achieve a constant there. Take an ant, far smaller than a man, but able to lift many times his own weight because the substance of his muscles is relatively tougher!"

"Well, power for polishing comes first, doesn't it?" I said, and we went to work.

The days went. Tediously our work progressed. Another spring came, and we lived two lives, with almost two sets of identities. There were classes and friends, and walks around the campus for Jan and me. Then we were

down again, where flecks of lint floating in the air looked like twisted twigs, and where metal surfaces were difficult indeed to burnish.

One evening, with grim excitement in his voice, Doc gave us some news:

"Again the Government is asking us for a favor. Small space ship and everything provided! There have been more mysterious breakdowns of equipment, and strange illnesses reported. So we're going out to Ganymede! Sorry, check out of the U; get your gear together, tighten your belts—because this is it! See a justice, maybe, if that's in your minds. Take a few days off."

My hand tightened protectively on Jan's shoulder. Somehow, before the unknown, I felt that marriage would be like a shield for us both, though we were still pretty young.

"Time to get hitched, Jan," I said later. "If you've made up your mind. Or should we consult George?"

Her eyes twinkled with a flare of recklessness. She lowered her voice and mocked humorously: "'This is a question which I am not permitted to answer. Unghh!'"

For a few days we were away from Doc Lanvin. Our brief honeymoon was on the Moon with my folks. Jan's parents had died in a rocket crash years before. It was good to see Mom and Dad and the old house again. Out beyond the sprawling buildings of the expanded labs, the skeleton of a huge hull was taking form. The stars, that meant, though the problem of overdrive, speeds greater than light, promised no early answer after all. A journey to the nearest stars would take a long, long time.

Within a week Jan and I were back on Earth, packing equipment. Armand Cope, whom I've mentioned, was one of those who came to bid us good-bye. His cynical mouth twisted as if he were both sorry for, and contemptuous of us.

"Maybe I was just born too late," he said. "I don't know just what you're after, but there are rumors. Well—honestly—keep safe."

"Thanks, Cope," I said.

"Yes, from me, too," Jan added gently. Then she threw a mild jibe at him: "Still trying to hold back tomorrow, Cope?"

Two guys I knew slightly, Bowhart and Scharber, were selected as crewmen for our ship, the *Intruder*.

"Bow and I are trained for simple space stuff, Charlie," Scharber said. "We won't stick our noses into this micro-robot business."

Scharber was broad and easy going. Bowhart was short and dark and serious.

Jan changed to the rough coverall, acceleration suit, and boots of a space wanderer. Maybe there was a regret for the difference, but it also brought a new jauntiness.

On a Sunday night our ship blasted off from the New Mexico desert. When our acceleration was completed, our ringlike hull began to rotate, to give us a centrifugal substitute for gravity. The outer silence closed in, and two months of monotonous journeying provided only a new setting for our efforts to build us metal forms that could stand beside an inscription on a sand-grain meteor as a man stands beside a monument. All of Doc's home workshop had been transferred to the *Intruder*. There, in the lab compartment, Doc, Jan and I sat hour after hour, wearing our control hoods, but living in metal bodies half an inch high that bent intently over an even far finer and more difficult craftsmanship.

We passed Mars' orbit without seeing the new man-made airdome cities among the ruins. We saw nothing of the asteroid belt where fortunes were being made in metals from the heavy core of an exploded planet. Our quest had a different goal.

When we finished three super-micro-manipulators, we were better prepared to finish our tools and equipment to make parts. But our tedious job was less than half done when we arrived on Ganymede, cold and bleak, its tenuous atmosphere composed mostly of unbreathable methane gas.

Scharber brought us down on the landing stage of Port Hoverton. The settlement itself was under domes nearby.

And Jan said: "Hurdy-gurdies, Charlie! Hey, Doc! Scharber! Bow! Beer, music, games. A last fling, like the spacemen and miners! Let's have it!"

So we did for a few hours. Then we had us a good sleep. Then we found a guide. Boom Harlow, he called himself. Oldish, cheerful as a gravedigger.

"Sure I'll take you to where that little tool chest was found," he said. "If you stay, likely you'll never come back."

He blasted off with us for a thousand-mile jaunt in our ship, arcing above the stratified mists of half-congealed gases that hovered over the Ganymedean landscape, and after we had landed at his command, he pointed out stone structures that looked both very old and very odd.

"There you are!" he said through the helmet phones of our space suits. "Maybe the last camp of the last survivors of Planet X, came here millions of years ago, before their world went ffftt—before there were asteroids. But there's something here yet, I'm tellin' yuh! Now, after you pay me, I'll get my mono-wing out of your hold, and fly back to town, and I hope I won't stop long before I rocket back to Seattle. Keep alive if you can. So long."

Boom Harlow was gone, then, riding a jetted metal triangle high against the thin murk. The rest of us were left with the creeps and the wonder, and all the work we had to do.

There was an ancient shelter of glassy rock, metal-lined, and once sealed, for there was an airlock. A dried, age-blackened form, its claws clenched, its queer, vertical ribs sticking through the skin, was crumpled in one corner of the shelter. Around him was his gear—tools and weapons like those in museums. Perhaps no one had dared to take them, for a curse was supposed to be here.

But you could tell that he had been moved; and the even cones of dust on the floor showed that every bit of it had been carefully sifted, no doubt after prospector Jeffers had found that tiny box of glinting tools. He had shown the box to reliable people; but then, by later report, had somehow lost it.

The mummy was an Xian. With what sadness at the loss of a home world, had he awaited death under the bleak sky of Ganymede?

None of us spoke. At last Jan kicked dust over the corpse in a gesture of kindness.

"It may be safer in the ship," I offered.

Scharber and Bowhart had the *Intruder*, and the simpler wants of the rest of us to look after. Doc, Jan, and I kept working on the smaller micro-robots. But during an occasional spare hour we'd send our half-inch alter-bodies out into the cold to look around the ancient camp. We found nothing of interest except a splinter of diamond set in a metal shank. A cutting tool?

But within a month, certain phenomena began to appear inside the ship itself. As humans, we couldn't have noticed. But being half an inch high, it was different.

"Whispers in dark corners, and shadows that move," Jan said. "And tinkles. Or are spatial solitudes affecting my mind?"

We had just put down her control hood, so she was speaking as strictly herself. She looked alert and curious, not overwrought.

"I thought I noticed something vague, too," I admitted. "Funny, the ship is sealed. Even the air is constantly being filtered. But who's complaining if there are developments so soon?"

"It would be best even to sleep in our armor," Doc advised. "Scharber and Bowhart, too. Yes, I thought I heard a muttering and chirping. And just now, as we worked on that final assembly, I was sure I saw something hide in a bit of floss that was adrift in the air. But they never come too close; they just hover near and wait. And only miniature microphones, like those of our robots, could ever pick up the sounds they make."

My hide was coldly atingle again. "Darn!" I laughed. "Why does the idea of smallness and beauty always suggest fairyland, unreality? Small things are just as factual as large ones."

"Fairyland is a dream, Charlie," Jan chuckled. "Something which many of us in humdrum surroundings wish was so. But a yearning can sometimes be made to come true."

Though we felt that we were being watched, nothing came to interrupt our hard work. At last we won our fight with materials when we arrived at the microscopic size-level limit of the workability of metals. We now had three true micro-robots. They were like their half-inch creators, except that they had two coordinated sets of eyes—a lensed pair to see by ordinary light, and another pair, fitted with magnetic focusing rings, to see by the rebounding of electrons from objects at close range, where detail was less than the span of a single light-wave.

Step two, down into The Small, was made like step one. We used the same control hoods, adjusted slightly, while we sat at our work table in the shop aboard the ship.

"To the bottom of the pit!" were Jan's final words as a woman.

We put on our hoods and plunged. It seemed infinite, this time. The gleaming walls and girders of the shop appeared as distant as planets. The surface on which we sprawled became pitted and scored from our new viewpoint. Polish was gone with magnification. An eerie, elfin ringing—perhaps the finer overtones of normal sounds—reached our tiny, tympanic ears.

We arose unsteadily. Our mechanical fingers joined, till we were a chain of three, moving toward the door of the box as a group. Then we were out on the undulating porcelain expanse of the table top. An air current, magnified to a hurricane by our minuteness, lifted us up till we floated free, still clutching each other's hands.

One peculiar thing about difference in size, is that the smaller an object is, the larger is its exposed surface in proportion to volume and mass. That greater surface in relation to weight, allows the bombardment of passing air molecules to lift anything of dust-grain dimensions and density into wingless flight. It also can give a sense of helplessness, as if the atmosphere has become a treacherous medium full of irresistible currents.

We tumbled, we laughed, and would have been panic stricken except for knowing that our real selves were in normal circumstances. Nearby, the air seemed to shimmer. A gnarled thing floated close—floss, looking like a twisted tree-stump, to which clear ovoids clung—some common form of microscopic life. A chunk of mineral dust came drifting nearer, its sheared-

off side glinting like quartz strata. Our two pairs of eyes still were not developed to distinguish colors. Yet Jan had reasons when she exclaimed in tinkling tones:

"Beautiful, truly beautiful! We came—we got here! In a sense, it's farther than the stars! But now what happens? Where are—*they*?"

"I don't believe they'll be long in coming," Dr. Lanvin said at last. "To write, to make tools, and to get into our sealed ship requires a capacity to think and plan. So, about us, they must be following a set purpose."

IV

Tension mounted in me. As we drifted in the air, I looked at our human selves, seated giants in armor, cowled, brooding, and of legendary height. Here was a chance for a meeting with entities of another shape, flesh, and history. For the Martians and Xians seemed as extinct as the dinosaurs. Their artifacts and mummies were known; but their voices, movements, and real selves, were elusively beyond imagining.

In most of the old imaginative stories of the future, beings from another region spoke and thought like men. But a recent University course had pointed out how deeply different must be races sprung from wholly separate chains of evolution, not only in form but psychology; how there would be no hellos or similarity of custom on the other side, and how one must wait with perfect self-control and mind utterly open, until an equal horror of alienness lessened in the alien beings, too....

Jan said, "Look." The word was a single, flat, undramatic note. But we saw them. A mass of lint, gray to our colorblind vision, drifted toward us like twisted branches. Out of it, as from shrubbery, a dozen pair of eyes peered—lenses with a moist glint, fuzzed at the edges; here I thought not so much of lashes as of strange, misplaced antennae. The creatures were like rough-hewn dolls, with craggy, almost triangular heads. Yet these were not metal robots. Their skin was rough, as from a coarse binding of spherical cells, still small, yet almost large enough to be seen individually.

These beings possessed two arms and two legs. Yet, in still another way they were familiar. They took all their major details from the mummied bodies of the Xians, though those original Xians had been of human size. What strange retreat, or advance, was implied here?

I was trying to answer everything about tremendous mysteries at once. But I heard Jan tinkle out words matching my own awe:

"Charlie.... Doc.... Other intelligent beings.... Real.... See their clothing, and the metal devices at their belts and in their grasp. Seeing something completely hidden previously, is getting closer to the Ultimate Secret of the universe, isn't it?"

The little robot that represented Doc, holding onto the right hand of Jan's proxy as I clutched its left, had things to say, too, as we floated free, waiting for whatever would happen:

"Critters as little as these micro-robots of ours—and intelligent, and of flesh. But there couldn't be an intelligent brain working on the familiar human principle in so small a size. The molecules are simply too coarse to achieve such compactness. For that and other reasons, these strangers have to have flesh of some advanced form of protoplast with its possible flow of many types of energy, submolecular or electronic. This might well apply to brain function, making its countless patterns not inconceivable in an almost infinitely smaller package."

Just for a second, Doc paused, before he brought his topic to an avid point: "Androids," he said. "Micro-androids, or the equivalent, in relation to beings not human! Is that what they are? Then it is another demonstration of the advantages of this improved, lab-developed basis for life—venturing into space unprotected, being almost indestructible—even going down into The Small!... Or could it have evolved naturally?"

The chill in my mind sharpened and turned more eager. But now Dr. Lanvin's groping words faded out.

For, warily at first, our opposite numbers in this strangest of historic meetings, at last went into action. As a group, each with a purpose, they leapt from that floating mass of floss, their graceful, swimming motions in the air aided by the reaction of hot flickers from little jet-tubes they carried.

Swiftly, as if taking a citadel, they surrounded us, and held us in their explorative, yet strong clutch. Now was the moment of blundering, of attempted communication, across the great, mysterious gulf of difference.

Doc addressed them: "Now what do I do or say? Who learns whose way of conversing? Or would it be trite to think that you might be telepathic?"

Could these beings even recognize Doc's friendliness? Well, we were in for a surprise. They had a spokesman. Out of his thorax came a slurred buzzing, struggling to mimic human speech:

"Telepathy? No, Mister. Not so good for us with you people. Funny? Maybe.... Learn conversing? One can always learn more.... But we have been visiting Earth, mostly unnoticed—since—before—there were—men."

Here was English, idiomatic to the point of slang. Yet, to add an eeriness, there were pauses, as if the effort to think in a human manner was more difficult for this trained but outworldly psychology than the speech itself!

So, the simplicity of communication was like in some of the old, imaginative stories. Well, why not, if these little people had been haunting human stamping grounds for ages? Besides, could extra-terrestrial thought, dealing with common physical facts, be so totally different? That University course had exaggerated.

Doc cursed happily: "Dammit, things'll be easy, now!"

"Easy," came the cheerfully buzzed answer. But soon I suspected that a cheerful tone was pure mimicry of a human way, without, necessarily, a real, corresponding emotion. For now our escort gripped us roughly, and drew us along through the great gulf of air, using hand-held jet-tubes for propulsion.

Doc's shouted, "Hey, what goes on?" and my equivalent complaints, were ignored. Our escort broke in two, six of its members, including the leader, continuing to lift Doc, Jan, and me upward through the air inside our ship, the other six, bearing what looked like massive equipment, falling behind.

In the ceiling of our lab compartment there was a circle, still edged with the rough scale of a tool cutting with intense heat, and there was a hinged, circular door of metal. They had cut through the skin of our ship, and had installed an airlock, quite like our own variety in principle, yet so tiny that our human eyes had missed it entirely.

Helplessly we were drawn through it, and onward into the murky night of Ganymede, over which Jupiter and his other scattered moons held sway. Our robot-selves of course did not feel the cold, which approached absolute zero. Nor apparently did our unarmored hosts. Nor did they seem compelled to breath oxygen.

"Charlie! Doc!" I heard Jan call. "I hope they're not taking us too far! The radio control hoods, keeping us in contact with our micro-robots, here, are of limited range. We could lose the robots!..."

"Not very far," came the answer from the being who had spoken before. "But Kobolah—myself—says it makes no difference to you."

Perhaps that strange little monster meant to reassure us. By now I had him identified as an individual. The irregular filaments around his eyes were longer and paler than those of his henchmen.

A Ganymedean wind wafted us along, our escort perhaps using it to cover distance, righting it only as much as necessary, with their spitting jet-tubes. Our course turned downward into the shadows of knotty rock masses near the old Xian camp.

We went through another airlock, and into a tapered, cylindrical chamber. Figures like the others were there, craggy, yet obliquely charming in form. There was what must have been a propulsive mechanism, perhaps refined by ages of development, until matter was totally converted to energy.

And there was a crystal vat in which complicated grids were suspended in gelatin. Deep in the menisculous, pearly medium were shapes, hardly seen, though suggestive.

Kobolah spoke again: "You three even built small robots with great pains to pay us a visit. So we thought that maybe you should truly come. We shall see...."

I saw that odd, triangular head. I could fathom nothing from the eyes, except perhaps a cold interest. But I felt tricked and trapped. As far as our senses were concerned, we were here, not back in our ship. Forgetting that, we had been off-guard there!

Can a robot have a fearsome headache? Suddenly I had one. Dizziness and a blurring of consciousness was followed by panic. Suddenly I was back in the *Intruder*, frantically unfastening the helmet of my space armor, then casting off the control hood.

I staggered erect. Dr. Shane Lanvin grunted beside me. His usually mild face was contorted. Jan gave a thick cry, her gloved hand on her brow. Doc and she had also torn off their helmets and hoods.

I floundered to Jan, heard her say, "Charlie...."

Then I saw a hole, like a tiny cigarette burn, at the fabric-and-wire elbow joint of my armor's left arm.

"Scharber! Bowhart!" I yelled. It was a thin wheeze. I wished that they knew more about Doc's work so they could help us. My final awareness was of the rush of their footsteps.

Time became timeless. Then I had a sense of struggling upward toward light. The effort was mental. A minute might have passed, or a year. I had a body which seemed to turn lightly on a mattress of coarse sticks. I felt like myself, clothed in real flesh. The light around me might have been diffused sunshine, and I saw colors, the familiar ones, plus what might be the indescribable paleness of ultra-violet, unknown to man as himself, and another nameless hue that perhaps was the sensory effect of electronic vision.

I didn't fully guess all this at once; but its ghost was in the back of my mind, and at the edge of panic.

I had sat up easily. I realized that I was still in the region of The Small. Once experiencing that environment denies any failure to recognize it later. Oh, there was the roughness of the glassy walls of the room, pleasingly decorated with geometric patterns like those of old tiles brought back to Earth from the asteroid belt. But I refer more to the insecure sense of buoyancy, of ease with which one might float in the air or recline upon it, after a tiny push at the floor. It is a feeling quite apart from the

weightlessness experienced in space; and though there was certainly very little gravity here, too, the difference remained palpable. And now I even felt a tingling in my skin—the impact of molecules, perhaps, as they tried to lift and carry me away.

My body seemed to conform to such a dimensional plane. It was me with some details blurred or omitted. I was clad in stiff imitations of the slacks and shirt I had worn inside my space armor. My hands, rough in texture, lacked the fine hairs, as if they had been left out in a process of transformation. Was the stiff, wirelike hair on my head still black? I fumbled at my face. The nose, large jaw, and brow, seemed the same, except for a certain shortness and roundness, as in a doll-like simulacra. Corresponding to this was the length of lashes around my eyes—or had electronic sense-organs been added, necessary here for close vision?

Again I looked around the room. One wall was absent. But the square left for ventilation was crossed by interwoven diagonals—bars which must have been incredibly fine wire from another viewpoint.

Beyond this barrier was an egg-shaped chamber, so huge to my present minuteness that it was like a mountain valley, its sides curving up in shade and lushness; though through its vitreous, natural roof, light streamed. Everywhere, bright green foliage peeped over garden walls. Sometimes it was shaggy and filamented, sometimes massy and spheroidal on thin stalks. Along streets rising in angular charm, were geometric masses in pastel tints, some unknown to man, before. There were cubes, pyramids, even spheres—buildings, obviously—yet of such simple oddity that a child might have designed them.

Water did not lie flat as in a lake, but gathered in great glistening dewdrops, burying a house or hill fantastically, but with startling beauty.

But all this moved with the daily life of a teeming civilization—living, manufacturing, buying and selling in the market place. The air was full of craggy shapes, some propelling themselves with arm and leg movement, others using jet rods. High on a slope there was a continuous electrical flicker, a bluish spark. Perhaps the furnace of a metallurgical process.

The springy stuff on which I sat, gross as brushwood, would have been cotton wool to normal-sized human touch. Perhaps it was vegetable fibre of that order. Crouching near me was a girl, clad in coarse blue fabric which in reality would have shamed our finest textiles. The details of her face were simplified in a doll-like blurring of line. But still she was recognizable, even with the lashlike filaments around her eyes.

Somehow I still spoke with my lips. "Jan." My voice seemed a miniature bass bell. I crept to her side.

Her courage and sense of humor were intact.

Her laughter was a tinier bell. "I'm all right, Charlie. At least, yet. Maybe I just don't realize. One thing we've talked about has happened, hasn't it? You look sort of cute, Charlie, like a puppet in a show. Doc, too." Jan laughed again.

Beyond her, dressed like myself, was the reduced image of Dr. Shane Lanvin, though his inner self remained unchanged, his triumphant smile just faintly edged with doubt.

"Hi, Doc!" I greeted. "Congratulations for success in a venture which began with you. Now, for the record, let's hear your version of just what has happened."

He smirked good-naturedly. "All right," he chuckled. "You can't get back to any control hoods, our former human-size selves. I've tried. So our whole identities must have been transferred to these far smaller forms. Somewhere in our adventures the structure of each of our brains must have been exhaustively charted, down to the finest wavering of cell-filament, and the least variation of chemical state. Thus must have been captured every phase of our minds, memories, and personalities. This might have been done by something analagous to our focused radar or X-ray photography, penetrating deep, and making an instant record. From this record, the pattern of our brains must have been rebuilt, with all the complex channels of association and so forth, but in a totally different medium, capable of a far finer and more compact flow of energy than mere nerve impulses. In a brain of protoplast, I think it could happen."

"Loose ends still dangle," I chuckled. "For instance, I remember a machine called George, and a statement by him that consciousness, awareness of

self, was even difficult to define. How about transferring that?"

Doc Lanvin shrugged. "Maybe the consciousness—the true self—is inherent in the brain channels, like the memory, and would also be transferred simply by copying them precisely," he said. "Or could the awareness be a kind of spark, capable of being captured and transported by an appropriate apparatus, as an electric spark can be captured in an electroscope? I don't know, Charlie. But I noticed some of the equipment carried by these Xians when they took us; and I thought of that."

Silence seemed to close in as Doc finished; and it grew heavy with monumental implications, almost apart from mentioned things. I breathed, which suggested that my present form was getting energy in the familiar way—by the combustion of food substances. But as I held my breath for a prolonged moment, there was only a brief flutter, as of a heart quickening its beat inside me. I wondered eerily if this was evidence of a casual change-over, as if my android flesh could so quickly convert to some other energy supply, perhaps that of radioactive salts naturally in its substance. Such minerals were fairly common on the Jovian moons, and far commoner among the asteroids.

I was compelled to breath again to speak.

"More could be remarked about, Doc," I said. "We know that the Xians were once of human-size, and of the same order of life. So somewhere in their long and checkered history, their survivors invented this new vital principle, and changed themselves. There may be various reasons why they chose to be tiny. Hiding, for instance. But as you once said, that's just part of the android advantage, and not the real issue. Here is a step in scientific development probably as much to be expected as television. If micro-androids can be made, so can larger ones! There's your pending problem on Earth, Doc, natural man versus his far tougher, more flexible competitor! Ultimate newness. It can be real! And wonderful! But to many it will be a fearful thing."

Doc's doll-like visage fairly shone. "The warning, eh, Charlie?" he chuckled. "The demigod dream coming to a head in eagerness and cold tension. Shock of the utterly novel versus tradition, even instinct! No ills; practical indestructability. Immortality, perhaps. The old, human hope! And

yet?... But should or can progress ever be stopped?... Damn, if we can only take this process of conversion home!"

"You two talk of going home, and of lots of big things," Jan complained. "But do we even know where we are? Just where is this room, and those houses and gardens out there, in a great hollow space like a bubble cavity in a glassy clinker? Of course such a cavity, a few inches across, would seem enormous to us."

Dr. Lanvin studied her soberly. "You're sharp, Jan," he said at last. "A bubble cavity, like in an old clinker. Umhm—m—many asteroids have that sort of structure, maybe formed by the sudden relief of a planet's internal pressure, when X was blown up. Steam and air made the bubbles in the molten, glassy lava. But when it cooled and solidified, the air, and the condensed water of the steam, remained sealed inside, unable to escape into space. Explorers have found microscopic green plantlife growing in many of those cavities, for through the glassy lava sunlight can penetrate, as it seems to do here. Thus, a perfect natural environment for living things in miniature was created. And a perfect retreat. By gosh, Jan, I believe you're right!"

Doc had always had almost a child's love for small objects. But my own enthusiasm was less complete. Call us all super-mites, placed beyond most of the physical ills of men; but Jan and I were still prey to nostalgia and panic and claustrophobia, for these are things of the mind. Hard men have gone mad in space, because they felt cut off from everything familiar. But at least they had their normal forms and size, and a known way back home. They weren't caught in a clinker cavity beyond a barrier of magnitudes that appeared more insurmountable than a hundred light-years of distance.

It was a treachery of our primitive thought patterns, I knew. It was against progress, and the explorative impulse. Yet I knew that it would have to be reckoned with.

Jan seemed about to answer Doc a little sadly. But then the grating over a circular doorway at one side of the room opened and Kobolah floated into our presence, and alighted before us. Uncertainly, Doc and I arose. No human yet could have read the expressions of Kobolah's queer, angular face, limpid filament-framed eyes, or palped mouth orifice. The ages of history, and alien thought structure behind that visage, were lost in enigma. But now his voice-tympanum buzzed; words came out with an effort, but their arrangement and apparent thought mimicked the human almost comically.

"Bubble cavities," he buzzed. "You are fine guessers. We are in a very small asteroid. But it is not in the asteroid belt. The great explosion long ago hurled it into an orbit around Ganymede. It is one of our many retreats. We wanted to conquer Mars. We attacked terribly. But they destroyed X. The few Martians still surviving tried to hunt our even smaller numbers down. But we found a way; we became little to be concealed. Later, we were at peace, safe. But being small was a habit not needing change. We bore offspring, as we could before. We built things up again, and multiplied, very few dying. We made more refuges in the solar system, then in the systems of the stars. We are strong and hidden. We have a good way. We are peaceful, except when there is danger. But you three have come—differently. All right, we can watch and learn from you, too. Yes, I have listened to all that you have said, but to learn is good, and not unkind. Right? Now I have answered some of your questions."

The buzzing voice ended in the slurred imitation of a laugh, which tautened whatever now served me as nerves. For to laugh is a specially human, Earthborn thing, not to be mocked. But here I was in the awesome dark of complete novelty.

Doc, however, gripped Kobolah's corresponding tactile member. "Does one do this, after all, among your people, Friend?" he asked. "Or express thanks? If so, here it is. As for the rest, about the technology of transformation—"

Doc did not even make it an apparent question. Yet the question was there. Dr. Shane Lanvin had to learn what he could.

Kobolah mocked up a human chuckle. But his monster's gaze was cold. "This is not for my decision," he buzzed. "But it could be as you wish. Yes,

I overheard what you want. Some I could show you now. You and your companions—Charlie, Jan. The apparatus you could see."

"Of course!" Doc replied quickly.

I looked at Jan. Her jaw was set grimly, as if to fight the strain in her eyes. I didn't have to ask her what it was. I felt it myself. All the strangeness around us, beating at, grinding at, our minds. Physical laws turned topsy-turvy, till nothing was the same. Could an android go mad—if the mind in it remained human, and reacted even against the unfamiliar substance of the arms and legs that it controlled? Too long already it had been so. We were realizing what we were. There needed to be some relief from the harsh thought.

"Wait!" I insisted. "Our own forms—are they dead?"

"Alive, sleeping, mindless, where they fell in your ship," Kobolah answered. "I believe—safe...."

My arm was around Jan. "There!" I said triumphantly. "That's better already, isn't it? You go with him, Doc. Jan and I need another mood, now. Ko-bo-lah—" I struggled to pronounce the name as he did. "Are we guests or prisoners? Can we go and come as we please?"

Finally he replied after what seemed an emotionless scrutiny: "I am chief of a project to observe you. Proceed as you like until stopped. There are common devices for propulsion there in the corner. The controls are easy. Have fun. Come along, Doc."

Dr. Lanvin took a proffered propulsion rod from our host. "Yeah—" he said a little dazedly. "Have fun. Be seein' yuh."

He still looked puzzled and amused as he followed the monster from the room. The grill of the circular door was left ajar. Down a passage beyond, daylight showed.

The little bell of Jan's laughter rang out, fringing hysteria. I patted her shoulder. "Easy, Honey," I urged.

She began to regain control. "Common expressions from a buzzing demon who might even be a good guy!" she said. "And around here you don't even walk, you glide through the air! Everything's crazy! And all the scientific explanations, while you get more and more homesick for your own self!

Darn it, Charlie, I'm a weak fool! But it's still all wonderful, beautiful! It should be enjoyed. That's the way to counteract fear and strain, isn't it, by enjoyment? No more deep theories for now! Let's go out there to the city, see the sights, follow our noses, try to have fun!"

"Right, Jan," I enthused. "Call us visitors in some exotic port. I guess we'll need practice using these jet rods."

In a moment we were out there in that lush, valley-like cavern, which really was a bubble, a few inches across, in the glassy crust of a fragmentary asteroid. The jet rods flamed and gave thrust in our hands as we maneuvered clumsily in the air, learning, hands joined to keep from being separated.

First we shot up to the immense roof through which sunlight streamed. Then we drove ourselves down over the gardens and towers of the city. Soon a curious crowd floated around us. They plucked at us, and their voices buzzed; but none of these Xians seemed to know our language.

"Does it really matter, Charlie?" Jan asked, her eyes beginning to shine, now, some of the strain already disappearing. "Here's an old, old civilization, hidden, grown esthetic, maybe even a little decadent, but extending far. You know it, feel it! Here are beings changed to an android life-basis so long ago that it seems natural—hardy flesh healing if injured, children being born as in the old flesh! Even death almost a myth! Gosh, I hope we can get used to all that, Charlie! Peoples multiplying, spreading to the stars."

"Don't paint it too bright, Jan," I laughed. "Come on. Let's explore farther."

I don't remember how many hours we spent on that long excursion, or all that we did. There was more than one bubble cavern; there must have been thousands connected by artificially drilled passages in double arrangement for traffic moving in two directions. In those passages, currents of air carried one along swiftly. It was a perfect transit method for a micro-world.

In some caverns were other cities. But there were more where tiny agricultural machines, with limbs like a beetle, crawled across miniature

fields. Here we ate strange, sweet fruit, that surely contained the carbohydrates of familiar food. But no doubt it also contained radioactive salts from the soil in which it grew. As we had been, it would have poisoned us. As we were, it was a double source of vital energy, chemical and subatomic.

Other caverns were murked with the fumes of electric foundries, self-operating, close to the mine-tunnels that bored deep into the natural, nickel-steel core of the asteroid. In still other caverns there were low buildings full of lathes, drills, presses, among those that we could name—all automatic, too. Then there were caverns where stood lines of square containers, enormous to our eyes, and joined by a network of cables. This must be a power source—banks of nuclear batteries.

And in several adjacent bubble cavities we saw where an enormous metal cylinder was being built, each oblong segment being welded into place by mechanisms of the true robot variety. From any one cavern only a small part of the curving side of the tube could be seen.

"Some kind of jet engine?" I asked almost rhetorically. "For their further expansion toward the stars? Like moving a whole planet to them, eh?"

"Your guess, there, can be mine, Charlie," Jan said.

We felt no physical tiredness in spite of all our activity. "Let's get back to a more idyllic surface bubble, Jan," I suggested, "and go swimming in water if natural law, here, allows it."

"Crazy!" she responded gleefully.

Air, rising in a vertical shaft, bore us aloft for the few feet that, to us, stretched into seeming miles. Against what appeared to be a green hillside, we soon found what we sought, a great, clear ovoid, glinting like a lens in diffused sunshine.

It almost proved true that we could not swim, here; for the relativity of smallness gave water a terrific surface-tension. It was difficult even to get wet! You could lunge at the dewdrop, and it would throw you back like a net of rubber. Even with android strength, we tried several times before we penetrated it. But then things went well.

Jan glided like a little pink nymph, silvery bubbles clinging to her face. We did not breathe. The greater relative viscosity of water did not trouble us. Our eyes did not need to close. Inside the dewdrop swam Xians who had followed us. And extending in crystal vistas were the furry green bulks of water algae.

Maybe there was no moment or place, yet, as beautiful as this. We enjoyed ourselves immensely. But grim questions about our future remained in my mind, though here and now the charm of fantastic difference reached a pinnacle.

"Now I'd like to go up and out on the surface of the asteroid, Jan," I said when we had emerged from the water. "The real test. Game?"

"Why not?" she answered.

So we found our way upward to a surface airlock. It's Xian guard did not stop us. The lock's mechanism was automatic. We crept out onto bleakness, with harsh space all around. Icy stars, silence, deep, dry cold. Huge Jupiter, gray-white, and streaked. The far-off but still dazzling sun. And blotting out a third of the sky by its nearness, Ganymede, murked by its moving surface mists, almost congealed.

"A test for the android—unprotected in the raw void," I said.

No sound came from my mouth; the vacuum made it impossible. Speech was purely a matter of lip reading, here.

But Jan nodded.

All I felt of an energy change-over was a protective tightening of my skin, and that quickened, momentary throbbing inside me. There was no sense of cold or suffocation, no pangs of blood boiling under the release of pressure. Perhaps our outer flesh now served as a sealing shell.

A sense of personal power came over me—android power. The thrill contradicted my darker dreads.

Somehow I wondered how much I had had to be redesigned inside. In any tiny body the relative viscosity of liquids imposes a definite strain on the heart. Were my blood vessels now made especially wide to reduce circulatory drag? I had heard that the littlest insects have to be somewhat special in their inner construction for this same reason.

More confidently, my mind reached out to all distance, and all unknowns. The demigod mood was on me.

It was then that a crowd of Xians emerged from the airlock. Horny digits clutched us.

We were drawn back into the interior of the asteroid, where the hoarded warmth of the sun was augmented by the decay of radioactive minerals. The crowd buzzed around Jan and me. Through tunnel and shaft we were guided back to the cavern and house of our first arrival, mistily illuminated, now that night had fallen.

Dr. Lanvin and Kobolah met us. Doc looked excited.

"Well, Charlie and Jan," he said, "I've met the real ruling force of this world, and have made my appeal. Come along for the answer!"

Kobolah led the way down a shaft that must have reached the center of the asteroid, the most protected place. Here there was a cylindrical chamber, the native nickel-steel of its walls gleaming silvery in the bluish fluorescence. Aerially, and on the floor, the chamber was crowded.

I looked up at a globe mounted on a spindle that traversed the central axis of that great round room. It gave off a faint blue glow. Its surface showed thousands of facets; but it was not rigid like a crystal. In its translucent milky mass were countless dark veins that pulsed.

"Think of George," Doc said softly. "The same thing in purpose, only far more so. Not a ruler, only an adviser whose opinion the populace respects more than its own. This is a great organized lump of androidal brain tissue of the same order as the condensed stuff now in our heads, according to Kobolah. It has the same volume efficiency, though millions of times larger. And it has all of the knowledge of this far scattered civilization at its command."

Jan smiled. "Poor old George," she mused. "I used to feel that his room over the library felt like a temple to Everything. Well, we've seen a few more mysteries, haven't we? And the feeling is here now."

There was a dry rustle in that steel chamber. First the message came in Xian. Then in English:

"Generally, the technologies of the peoples throughout the cosmos will achieve a sounder, more lasting state of the body as soon, or sooner, than it is deserved, and can be handled intelligently. When it is new, often there is fear, confusion and sometimes disaster. On Earth, the native invention of a process of this sort cannot be more than a century off. In each case it should come at about the time of the first journeys to the stars. But the perfected invention, as it exists here, is better than a crude beginning, which will add to danger. Essentially, Earthians are about as ready emotionally as they will be in a short hundred years. The universe seeks to improve its awareness as rapidly as it can. There will be danger; this is a warning. But it is recommended that the conversion method be demonstrated to the Earthians as a gift."

The rustling voice clicked off.

"Thank you," Doc said solemnly, his gaze directed upward at the great globe. "Thank you, too, for pointing out risks."

Then he turned toward Jan and me. "Yes," he said, "Kobolah tells me that it has a consciousness, unlike old George. And I'll take a chance, in spite of a man at a fire, fuddled in a world changing too fast for him. Anyway, what else can we do? Scientists can't stop studying and learning any more than they can stop breathing."

Kobolah's filamented eyelids blinked. "Then come," he said.

We reached the labs where our intensive instruction, which was to last more than an Earth-month, began. There we found our three micro-robot bodies of metal, kept as in a museum. In other rooms were the furnaces, subjecting silica, hydrogen, and other chemicals to great pressure and heat.

We became acquainted with the vats in which readied substances were held in solution. Next, under Kobolah and Nintan, his superior, we studied the shaper grids and power sources, and the intricate regulating devices attached.

Finally, an insect-like animal of natural protoplasm, native to these bubble caverns, was made the subject of a demonstration. He was bigger than we were, and tolerant to the radioactive poisons of his environment. Otherwise, he was of the same vital principle as humans.

Anesthetized, he was immersed in a gelatin-like solution. Power flowed. Slowly, the substance and chemistry of his tissues was altered, cell by cell, without change of form, and never losing the inner motion of living. It was a process remotely akin to electrolysis.

This was the simplest change that could happen. But there were others. A body, or its three-dimensional simulacra made in any size, could be used as a pattern for a protoplastic form, and made to grow in another vat. But necessary alterations could be interjected too.

The nature of consciousness remained obscure to me, even under instruction. But the idea of a special indivisible spark or node of energy seemed to remain an at least tolerable analogy. Doc Lanvin's comprehension here was a lot better than mine.

"About the awareness, the philosophers were almost right, Charlie and Jan," he said one day. "But science can touch it too, reverently, as it touches a beating heart, which is a pump, easily understandable by physical law. So it is with the awareness, too. Who would want it any different? Who would want the soul to be merely a formless miracle of command, when Divinity must be logic and order, and completeness of understanding?"

VI

Somewhere along the way, this and other matters became too profound for me. I absorbed what I could; but my field is action and feeling, not deep penetration, like Dr. Lanvin's. He pursued androidal conversion down to its last secret. Drawings and formulae, changed to Earthly terms, went down on parchment, and into his head. He toyed with the wondrous slimes of another kind of life, and at last understood them.

Jan and I were lesser beings. Buffaloed and a little dazed, we would wander off from the labs. Often we swam and laughed. Part of our personalities was adjusting to the fantastic region of The Small. But we worried, too. About our original bodies, and about a reticence before questioning, on the part of even Kobolah. Then Jan expressed another thing:

"Have I learned to read suspicion in the manner of the local folks, Charlie? Their minds are beyond us. But to them, recently, we have been strange giants beyond easy imagining. Now, do they especially resent having their greatest secret given to us? Do they object to the advice of their version of George, that we should have it? I feel a danger, Charlie. They could destroy us, or keep us here. Already they won't let us go up to the surface of the asteroid, though gosh knows what we could do there."

Jan and I were crouching in a little glade, in a lush cavern where the sun shone. No one else was near. I said softly:

"From the surface, I think we might get back to Ganymede and the *Intruder*, and maybe to ourselves, if it's not too late."

Jan looked at me with a wondering frown. "Yes, she mused. A few inches is a mile to us. Some ways, our movements are terribly limited. But in other respects, we're more free. With only a jet rod, we might travel those thousands of miles."

"It's an idea to keep in reserve," I said. "But there's another trouble. We've been here for about two months—counting one for the changing of our forms. Would our own bodies, even if they are still alive, or our ship and Bowhart and Scharber, still be where they were, after so long?"

A trapped, icy feeling came over me.

Jan was a real pal. You didn't have to hide your fears from her. She was a courageous realist. Her little rounded face only looked sort of stern.

"What to do, Charlie?" she answered. "Wait and see, I guess. Funny how important old familiar circumstances are. But we'll get along—even always being what we are, now. Darn Doc, though, never thinking about anything but his studying. Double-darn our Xian sponsor, Kobolah! Hint about our personal futures, lately, and he gets as elusive as all the history of his kind!"

I chuckled bitterly, and then quoted some of the things Kobolah had buzzed at us: "'Leaving soon? How soon is soon? To a long life, a century is nothing. Are you not happy?...' Yeah, that's Kobolah! A demoniac cross between something we'll never quite understand, and a kid denying with naive aplomb that he stole the cookies."

Yes, an elusive inertia of suspicion was all around us now, like a barrier.

Jan and I got through to Doc Lanvin at last, penetrating his studious fog. An overtone of grimness came into his mild expression.

"I've noticed the change in Xian attitude, too," he admitted. "It's a shame to be wanting to skip out on them, now that I've learned all that is necessary. But with the biggest piece of potential human history in my possession, I could hardly let minor qualms deter me much, could I? We'll find a road to freedom."

Yet it turned out less easy than Doc hoped. Time after time we approached various surface airlocks. Redoubled Xian guard-groups pushed us back gently. Neither stealth nor violence had any chance of being effective. We were constantly watched and outnumbered. Twice we tried hiding in metal boxes, full of parts destined for the surface-assemblies of the tiny world's slowly developing star motors. Both times we were promptly discovered, and pulled forth with emphasis. Xian voices buzzed. Their eyes were cold. After that second try, Doc had a wild look, like somebody with a treasure that he can't use.

"No star trips for us, yet," he growled. "Not with another bigger purpose back home. Somehow I'll get there, or stop living!"

A little later we were back in the familiar laboratories. It was night, deepened by the fact that the sun was now eclipsed by Ganymede. But in the windowless lab with its electron lamps, this couldn't matter. Kobolah pattered in a corner. No one else was with us.

Keyed up and angry inside, I noticed a rather unobtrusive combination of circumstances—three new jet rods in a corner; small nets of fine wire, containing steel cylinders of supplies. Casually stuck to a metal prong on

the wall was a parchment map, showing a vertical shaft leading to an airlock—the lab's private exit. Beside the map, a little used grille was slightly ajar.

Excitement became a kind of panic inside me. I looked at Jan. Her long lashes blinked knowingly. Doc nodded and walked casually away. The parchments of the secret he had gained were nearby. As if only to add further notes, he took the vast sheaf out of its compartment and carefully divided it into three. Midway in this operation, Kobolah turned toward us. Millions of years of difference in background, and in physical, mental, and emotional form, looked at us from great, cold eyes. A nervous chill came over me, both from the bleakness of discovery and frustration, again, and from the namelessness of that gaze.

Finally the monster imitated a harsh laugh. "Call this outburst peculiar," he buzzed. "Coming from nothing. But I happened to think that it is easy to be a fool, and often one will never know which way is foolish. Remember that."

He turned his attention back to the sputtering electrical apparatus over which he had been working.

"Thank you, Kobolah," Jan said nervously. He did not reply.

We divided the parchment among us, gathered up the equipment, and slipped quietly past the exit grille. An air current lifted us up the shaft to an unguarded airlock, whose control devices were readily responsive.

"Somebody stacked the deck for us," Doc whispered. "The scientist's logic, against popular doubts, maybe? Better to let us escape, than to release us openly, eh? I hope he doesn't get into trouble with his people. Or is there a deeper trick? Well, we'll soon know."

We emerged onto the deserted surface. We were micro-androids in space; dust-grain things matched against the universe, and the future of man. But we were part of both.

In the shadow of the asteroid world's eclipse by Ganymede, there was still soft light from Jupiter. Now we joined ourselves like mountain climbers, with a thoughtfully provided floss cable. Then, with small bursts from our jet-tubes, we leapt.

Soon we were falling toward Ganymede, accelerated by its attraction. It was a trip of many hours. Our jet rods checked our speed while we were still in space, and the satellite's atmosphere became a supporting cushion. We had an advantage over full-sized people—we could not fall to destruction. Instead we had to search for downdrafts to help force our descent with the rods.

Completing our journey, however, was not especially difficult. In Ganymede's glowing crescent we located a foamy dot—the airdomes of Port Hoverton. From this reference point it was easy to determine where we had left the *Intruder*. We got down into a prevailing wind. Thereafter our progress was swift.

After a few more hours, and some jockeying with our jet rods, we knew we were over the right place. We could speak audibly again, now.

Doc's grin was a bit forced. "You can even see its circular imprint in the dust," he said. "But the *Intruder* is gone."

Jan pointed below. "There's a space tent, Charlie!" she exclaimed. "The little brown dot! See? And somebody's standing before it!"

Swiftly we jetted down toward that bulging, inflated tent, fitted with its zippered airlock compartment. It stood alone in frigid desolation. "S.S. *Intruder*" was lettered on its side.

We alighted on the plastic face window of the armored figure, and clung to scratches in the material.

From this position we looked at the face of the man, huge, handsome to our former view, but made ugly by magnification. The skin-pores were craters. Individual scales of the epidermis, with the living cells beneath, were all visible, on forehead and nose, and around the colossal eyes, in which the separate flecks of pigmentation could be seen. It was an impressive, belittling vista.

The colossal jaw worked slightly: the narrowed gaze looked grim.

"It's Scharber!" Jan said. "He stayed here to keep watch, hoping for a sign from us, I'll bet! He knew part of what we were doing. But now he doesn't even notice us, any more than you notice motes on a windowpane. And how

can we talk to him? He could never hear our voices directly. How can we get anything across?"

The riddle faced us tautly, as if we were trapped forever in a lesser dimension, even beyond communication with our own kind.

"The jet rods again!" Doc shouted. "He'll see the spark of blue fire!"

Doc braced himself in a scratch ridge in the plastic and squeezed the trigger of his rod. At a little distance, the glassy surface boiled up in dazzling flame. When the thread of intense atomic heat was broken off, a smoldering pit was left in the outer surface of Scharber's face window. A mere pinprick.

But plainly Scharber had observed, and added it up. His great eyes widened; the plateaux that were his cheeks, paled. In the canyon-like ridges of his brow, came the sweat of fear. Drops of it were bulging lakes, rushing down past the lopped-off redwood trunks of the blond bristle along his jowls.

It was then that I found that another's fear of the unknown can inspire fear—which was easy to feel, anyway, when looking up at that mighty visage. Here was I, minute before this Atlas. I felt outclassed beyond measure.

There came suddenly a great shock of sound. Almost, it was more a heavy vibration, like an earthquake. Quivering with it, Jan, Doc and I clung to the roughness of Scharber's face window. Yet it had the beat of recognizable words. Scharber was speaking:

"So you've come, damn you, whoever you are! Like you came for some part of Lanvin and Charlie Harver and his wife. Well, their bodies, still in deep coma, were shipped back to Earth a week ago on the *Jovian*! We have scientists to figure out what you've done to my pals. Bowhart has gone to help the scientists with what we know! So look out! We're strong on Earth. We can fight and punish. So—to hell with you!"

Scharber was terrified before the unknown, but defiant and brave. The oldest human virtue was there, and it gave me a lift.

"I wish we could thank him for that kind of talk," Jan said.

"Maybe we can," Doc answered. "But our big problem is to get home fast, now. Ships from Ganymede to Earth run only every two months, and if the *Jovian* left only a week ago, there aren't any ships here! And how long before coma becomes death? When it has already gone on for so long? I know how you two must feel. With me, maybe it's not quite so bad. But darn, I still need that carcass of mine!"

I looked again at Scharber's frightened face. I had hoped that he could help us. But without space craft, that was unlikely. Oh, a call might be sent for a rescue vessel. But it would be sixty or so Earth-days in arriving, even if involved explanations of our peculiar position could be made by interworld radio.

"There's a way to communicate with Scharber," I said. "We could probably get him to have a message sent to Bowhart to recall the *Intruder*. But to turn a fully accelerated space ship around in mid-trajectory is no simple trick. Anyhow, there'd still be a bad delay."

"So, beyond trying to locate some small craft at Port Hoverton, there's just one other thing for us to do," Doc said grimly.

Jan expressed it for us: "Use the same method that we used to come here from the sub-moon of the Kobolah's people? Go without a ship at all? Achieve a high velocity; trust ourselves to something over four hundred million miles of empty void unprotected? Is that what you mean, Dr. Lanvin?"

Her small face looked pinched and awed.

"That's exactly what I mean," Doc replied. "As things are, I believe that it is considerably the best way. Oh, we can still die, I imagine, under certain circumstances! But the stakes are pretty big. I'd suggest that you stay behind, Jan, until we could send for you. But the form that was you is also one of those in a coma; and time is undoubtedly precious. Yes, there's desperation of at least a minor kind in what I suggest. But I think we've got all that we will need. And being as tiny as dust gives us certain definite advantages."

Jan looked at me soberly. "Sometimes small, inert objects actually leave worlds on their own, don't they, Charlie?" she mused. "Not only atmospheric molecules achieving escape velocity, but sometimes much

more massive particles? At least, there was the Arrhenius theory of the propagation of life throughout space—by means of spores torn from the upper air of one world by the light-pressure of its mother star, and propelled by the same force across the interstellar regions to the planets of other suns. About ourselves—well—aren't we about the right size and toughness to travel in approximately the same way?"

I looked at Jan, gulped hard, breathed "Okay," raggedly. Then I returned my attention to the enormity that was Scharber's face.

He had hardly moved; his eyes continued to search the curve of his face window, as if needing another sign from out of the unknown—as if, in fascination, he feared to miss such a sign. But his sweat of terror, at least, was subsiding.

"He can help us, a little," I said. "But staying here, he's unhappy, and can't do any good."

Doc nodded.

"So we do the right thing," I chuckled. "First we change position; mount to the top of the metal flange that frames his face window. Just let go of this plastic surface, you two."

My jet rod flashed. The cable of floss which joined us all, drew Jan and Doc after me, as I shot outward through the air to the crest of the flange. There we clung. I had to hold on tight to resist the kick of the rod, which would have hurled us far out into the air again, as I used it for a pencil to etch a message on Scharber's face window with its long needle of atomic heat.

It was like writing on the sky. My arm swung wide. But the range of about half an inch, with the rod's energy roaring at full, was about right to give me a normal-sized script. The jet's kick was trying to break my arm, otherwise it wasn't too hard to do.

I even tried writing backwards so that Scharber could read my message normally from inside his helmet. Where the needle of heat touched, plastic seethed, and a visible line was left.

I wrote:

It's us, Scharber. Lanvin, the Harvers. Changed. Micro-androids now—like race of Xian origin. Friendly. Go home, Scharber. But

please send radiogram. Urge imperative need to keep our bodies alive. Will return to them. A million thanks for everything.

The eyes of the Atlas who was our friend, stared again. Lakes of nervous reaction came into them. The plains of the cheeks whitened, as with some strange frost.



The eyes of the Atlas who was our friend stared ... stared fearfully at the message being inscribed so mysteriously on his plastic helmet.

The earthquake spoke:

"Charlie? Or am I going space nuts? Maybe this could be.... But who ever heard of it!..."

Panic at his own thoughts made Scharber move suddenly. The movement threw Jan and Doc and me from the frame of his face window. As we tumbled in the thin, methane atmosphere of Ganymede, I heard Doc laugh.

"Scharber will probably be all right," he said. "It's the shock of difference, again. But the message won't vanish from the plastic. He won't think that madness made him dream it. He's tough and young. He'll straighten out.... Come on, let's get started—for Earth!"

VII

We were lifted upward toward the limits of the atmosphere by the jets of our rods, aided by natural updrafts, which we sought out. Joined together as a group by the floss cable, we were certainly far heavier than any of Arrhenius' theoretical spores; but we had the advantage of intelligence to seek out forces to help us. We were not inert particles to be buffeted by chance impulses of nature.

We attained the high ionosphere of Ganymede where the sky was almost as black as space, where accelerated residual molecules beat against us, giving us some of their upward motion, and where sound had almost been smothered by the vacuum. There, Doc clutched mine and Jan's hands for sonic contact, and said thinly:

"Last chance for vocal speech. We'd better know how we all feel. In the parchments we carry we have about the greatest possible material gift to man, a dream of his from his beginnings. Practical freedom from death, from physical affliction. Immensely increased range of possibilities. The universe, in almost all of its phases, can truly become his stamping ground, now. It's a treasure that men would kill to have. To me it is an inevitable and wonderful step of progress. But there's a confusion in it, based on a split in human nature. You've seen and felt how it works. The mistrust of old

instincts for the completely different and revolutionary. Fear and even horror that invokes a savage compulsion to fight back. There's trouble ahead—between the two halves of man's character—represented by eagerness and revulsion. We know how it is from our own feelings. The android bodies we now have are the substance of the treasure, the gift. We exult at its legendary advantages; yet we have a terror of a strange exile, if we can't get back to our weak, natural flesh! The answer, on Earth, when the Big Change really comes, will be emotional adjustment, acclimation, time. Right?"

I answered Doc quickly: "You're looking for possible treachery in our nerves—opposition. Don't worry, Doc. I know every feeling you've mentioned. But the balance is all on the side of belief that progress is inevitable and good. I say this as a pretty average guy, Doc. Let Jan speak for herself."

My wife smiled. "Charlie knows I agree," she said. "So let's get the show on the road...."

We rose still higher in the atmosphere, to a place where the rising sun's rays were like a thin wind blowing against us. It was the pressure of light, the same thing that makes the vanes of an antique photometer spin. Gently, but with increasing speed, we were urged into space. One thing was wrong. Our Earthward course was to sunward, against that minute thrust of solar light. But there was a way to correct part of this.

Accelerating with the help of the pressure, we swept around Ganymede in an orbit; we waited until our direction reversed itself, as must always happen in circular motion. Then we really built up velocity with long bursts from our rods, and tangented sunward, breaking our last tie with the Jovian moon. We were on our way.

I felt my hide stiffening defensively. Over long periods we were not entirely without need of shelter in the awful spatial dryness, so we kept watch. The void is not completely empty. It contains many scattered hydrogen and helium atoms, and a rarer sprinkling of cosmic dust. We were lucky. Gleaming like a planet reflecting solar light, we saw a lump of rock moving with us toward the sun. We jetted to it and clung, laughing silently in the vacuum.

Doc's lips formed the words: "More speed. Time is short. Use up the cartridges of the rods. We have more."

Any object, broken clear of the gravity of a planet or large moon, is free in space. Acceleration is resisted then only by inertia. A relatively small force can build velocity enormously.

We were traveling at many miles per second when Doc mouthed: "Not too much. Eventually we must apply the brakes."

We fused our way into the meteor with our rods, and hollowed it out. We closed the exit with the slag of our excavations. True, the sun's radiations were a source of energy to our android tissues; but they also hastened drying—our worst enemy here since our body fluid was water.

As time went on, our skins hardened further, forming a kind of shell around the moisture in our vitals. And we had a small supply of water in steel cylinders. In a pinch, we needed little. We had food, too, similarly packed—Xian gelatins containing the radioactive and other minerals necessary to sustain protoplasmic flesh, and give it a sure energy source in space.

While we were burrowing into the meteor Jan did a whimsical thing. With a diamond-chip tool she inscribed over the entrance of our cave:

"Dr. Shane Lanvin and Charles and Janice Harver traveling to Earth in the Miniature—2037 A.D."

"There," she said in silent lip motion, for the reading of which we were gaining practice. "Maybe the inscription on that quartz-grain meteor you used to carry, Doc, was just as casual. Maybe it was carved, just on the spur of the moment, recording a journey of little Xians."

"You happen to be right," Doc answered. "I knew those hieroglyphics by heart. I drew them once for Kobolah. He translated. Four micro-Xians were traveling the short distance from one of their inhabited asteroids to another."

Later, the three of us fell into a kind of sleep. Or was it creeping death? It scared me. Our metabolism slowed. Consciousness left us. And so, time went very quickly. Maybe our tissues actually froze. I know now that this hibernation is a natural android function, conserving physical forces during long periods of inactivity. And it could not stop for good our rugged vitality.

We revived when the sun was nearer, and warmed us more. Stiff with dryness, we drank water, and loosened up our muscles.

We cleared the exit of our burrow, and crept out on the surface of the meteor. Rushing on in its elongated eclipse around the sun, it had come close enough to Earth to make the latter a disc of about one-quarter the apparent diameter of the Moon, as seen from Chicago on a clear night.

"Our meteor probably won't get much closer," I mouthed. "So we might as well jump soon. No use wasting the energy of our rods, decelerating a meteor mass, too."

Doc nodded.

"Where will we be most likely to find our old selves?" we read from Jan's lips.

"At the Space Medicine Research Hospital, near Chicago, I'd guess," Doc answered. "They send them nearest our homes. Or—peek over a shoulder at a newspaper, or into somebody's television. I think we are news. Are you both sure you know just what to do if that old protoplasm of ours hasn't got tired of waiting for us?"

"Yes," Jan replied.

"Fine," Doc commented. "So we'll drink some water, eat a little, limber up, and then start for home without the meteor."

Much of our physical forces had returned with the prospect of activity. Like any awakening, it was a natural tuning up of body. But I think that even our android chemistry had suffered in our vast journey. Doc and Jan both looked thinned down. I hugged Jan in appreciation of her unwavering spirits.

"Good kid," I said. "It shouldn't be long, now, with luck."

We all jumped, then, and broke our velocity in one direction with our rod-blasts, bending our course toward Earth, now only hours away, even at

steadily declining speed. And so, as unheralded as ghosts, but as significant as a new dawn of history, we came in.

Yes, we still hit the fringes of the atmosphere a bit too fast. The floss bond, holding us joined, burned in the heat of friction. Thereafter, there was no keeping together in tumultuous vastness, that, though it was just the Earth's air, seemed infinite to our tininess. I could cry out for Doc, and more especially for Jan; but there could be no answer. Really, it was the first bad break we had had.

I was high over a coastline. And a circumstance, particularly effective in The Small, helped me to orient myself. I found a bit of quartz-dust floating near me. I clung to it. Yes, I had heard of quartz crystals functioning sometimes as natural radio receivers. But my tiny ears were much better designed than the human to pick up minute sounds.

For more than an hour I listened to overlapping broadcasts. But the most powerful station I heard was in Frisco. So that was the city beneath me. I heard several newscasts. Parts of them were significant:

"... Dr. Shane Lanvin, micrologist, and the Harver couple, his associates, seem near death now in Chicago. For almost five months a spark of life has been sustained by intravenous feeding and other therapy. Dr. Lanvin's party was sent to investigate certain threatening micro-phenomena in the vicinity of Jupiter. Should any credence be given to a fantastic radiogram sent from Ganymede by another member of the party about a micro-race of supermen? Perhaps not; but it has been the thing that sparked the special effort to sustain life in these three during the past six weeks."

I was already jetting, riding the prevailing winds high in the stratosphere, and at last grabbing a lift on the skin of a passenger rocket-plane. From high up Chicago looked almost as it did from normal human eyes. There was no feeling of being lost in enormity, at least. That was how I found the Experimental Hospital, and descended toward it. The rest was easy. I had only to follow the newscast men to the three rooms.

Hovering in the air, I felt the thunderous vibration of a doctor explaining wearily for perhaps the thousandth time:

"Tissues and organs have no fundamental defect; some repair and replacement has even been made. There should be consciousness, but there is not. The rest is mystery."

I went to Jan's room first. How long had it been since I had seen her real face? It was waxen, now. All the color faded, as in an old painting. Never mind how I felt; it was bad enough. I drifted to Doc's room. His eyes and cheeks were sunken. Hovering high over him, I could not tell that he breathed. Then I saw myself, gigantic and pallid. The embarrassment of seeing this corpselike thing was lessened by the fact that it resembled my former lusty self only slightly.

"Hurry back, Jan, please," I urged aloud, though no one could hear me. "Hurry back, Doc."

I heard things in that room. Physicians conversing in thunderous undertones: "I'm getting tired of this. Interesting case, but it has been too long. Can't last much longer. Yes, sometimes it seems an unkindness to try to maintain life in something doomed to die."

Now that there was a chance at last, the help of those doctors might be wavering.

I found an interne writing at a table in the corridor. It occurred to me that, had it been necessary, I would even have dunked my entire body in ink from his pen-nib, and written him a message by dragging myself across the paper of the form he was filling out. But I still had my jet rod, so I clung to his knuckle, and scribbled on the form in a charred line left by a needle of atomic fire:

I have returned. So have the others. Please continue your efforts. Thanks.

Charles Harver.

The interne's hand jerked. I was hurled toward the ceiling. But I heard his bone-jarring roar:

"Hey—Fletch! Dave! Look at this!"

If they didn't understand or believe, still they would be alert and interested. There would be no breakdown of their struggle to keep those bodies living.

I went back to the pallid thing that had been I, and did what was necessary, after I had cached the parchments I carried, and most of my equipment, in a groove in the molding on the wall. I allowed myself to be inhaled. Deep in the lungs, I cut my way into a capillary with a diamond splinter. It was an insignificant wound, really. Then, in a rushing flood, while dim, reddish light penetrated to my eyes, I was borne along. I knew by a violent turbulence that I passed through the heart. Then there was a sense of rising. Absolute gloom meant that I was inside the skull. There I lodged myself in as small and unimportant blood vessel as I could find.

The rest was simple after that. I merely relaxed. It seemed that I went to sleep. But I was in my own brain. Encouraged by a natural affinity, the little energy-node or whatever it was that was my awareness and my ego, went home. It was, shall we say, a wanderer's return.

When I awoke it was mid-morning. The mental pictures of recent events remained vivid, yet they had assumed almost the character of a dream. Beyond my window were maples and pines. A robin was scolding. It was very pleasant, indeed, until I thought of Jan and Doc.

"Mr. Harver, you're awake!" a nurse exclaimed. "We knew from last night's tests that you were suddenly much better! There had been a message written in an impossible way...." Here, the girl looked frightened.

"Never mind!" I growled. "How is my wife? And Dr. Lanvin?"

"Mrs. Harver is still asleep. But even her color is far better, and she smiles to herself. Dr. Lanvin is much improved, too, though he is still very weak, and has not regained consciousness."

I sighed with relief. They'd gotten back just as I had. Yet, with what we'd brought back, this was not an end but a tense and wonderful beginning. The android secret. Improved man, large or small. A revolutionary fact to be thrust on our mortal race, with all its doubts and enthusiasms and prejudices; to be pushed into the age-old familiar sequence of birth, death, happiness, suffering, and decay of our kind! It was monumental in its possibilities for triumph and disaster; and for a weak moment I had a mighty wish not to disturb the peace, and to let all of this sleep forever.

Of course doctors and newscast men talked to me that day:

"... The message? 'I have returned....' Just what, in plain language, did that mean?... What did you find in your explorations in miniature? There is a story from somebody named Scharber on the way to Earth from the Jovian system, now. A yarn about a race that made itself unbelievably small. Yes—to hide itself, I suppose."

"You might like the story when and if you hear all of it," I answered. "Let Dr. Lanvin, my superior, talk, when he is able."

Late that day I was on my feet briefly. I held my wife in my arms, saw her smile, heard her say: "Well, here we are, and what now, Charlie? I even wonder if folks will be disturbed to know that tiny Xians have been visiting Earth for ages, unnoticed. It's kind of creepy."

Doc grinned up at us wanly from his bed. "This carcass of mine seems pretty well spent from the strain of my absence," he laughed. "Oh, I guess the damned thing could be patched up some more. But why bother? When I can have another body, same size, same shape, same organs, including a brain duplicated to the last filament of a brain-cell—no special principle required, as in *The Small*—all built of tough protoplast, and with a few things straightened for a youthful appearance and advantages? Not a robot any more than a man is a robot, but a human of firmer flesh, capable of all that a human is capable of, but much more. Glad to see you two up and around."

Yeah, Doc had always been a progressive. Oh, he'd had his doubts, too; but now, if the Great Change fazed him at all, he didn't show it.

VIII

Jan and I soon left the hospital and set up housekeeping in an apartment of our own. But with all that medical science could do, Doc still had to stay in bed for a month. But he started directing the forces of destiny, almost as soon as he could give orders.

I was in on the deal, of course, as were several doctors from the hospital, and Bowhart, and Scharber when he arrived on Earth from Ganymede.

"Gonna do it, Doc, aren't you?" Scharber said, when he first saw him lying there, pale and wasted. "You lugs scared me plenty once. Now, though, I feel foolish. Big words you need for this! It's the dawn of the demigods!"

My blood thrilled with a mighty promise, too. At night, going to sleep, I'd exert my will. Lodged inside my head was a micro-android. I'd will myself into it again. And so, for a little while, I'd escape from my own mountainous form, to float free in the air and consult the notes and drawings on the parchment that I'd hidden on a molding in my hospital room.

Doc and Jan would do the same. They, too, following the plan we had made in space, had similarly cached their portions of his notes. But now we had assembled the complete record of the android process in Doc's house.

And so the beginning was made. When Doc was able to get around again, things really got under way. He obtained a government grant. A whole lab and a large staff of workers, was set aside for us. Retorts, pressure-vats, and other apparatus to produce the basic materials, were constructed and installed.

Ours was a major project, coinciding in time with another major project. For the first real starship was finally under construction on the Moon. Three more years it would take to be completed.

But our enterprise reached practical fruition in fourteen months. I was among the men present when Dr. Lanvin lowered himself into a tank of special gelatins.

He was nude and emaciated; yet he kept his humor, and a certain dignity. A thin hand made a slight gesture. To Scharber and me and the others, he said:

"This will be the easiest trick, learned among the micro-Xians. Simple tissue-replacement, cell by cell. Improved protoplast in place of protoplasm. That's all. Well, wish me luck."

The anesthetic that had been injected into his veins worked. He slumped down gently. The gelatins closed in over his face, and the month of slow

gestation toward rebirth began. I saw his body at various stages of the process; little changed in appearance except for much increased robustness.

Other duties intervened, so I did not observe his actual removal from the tank nor his reawakening. But Jan and I met him a few hours later, as he left the small hospital of our lab. The old gray suit he wore, hardly fitted him. He still had his ragged blond mustache. You could tell that he was he—with many years subtracted. He looked about as old as I was—twenty-three. But these were the only signs.

He grinned like a kid, jubilant, but a bit self-conscious. He said, half joshing:

"Look me over—the miracle of the era, the successor to natural man; and no casual observer could ever tell that I'm not as humans have always been. I eat, I breathe oxygen; I need some foods with a different mineral content, it is true. I sleep if I want to. Given a mate of like substance, I can reproduce my own kind. But I won't age. Cut a finger off me, and it would manage to live independently for a long time. Wound me terribly, and I'd probably manage to heal up somehow. Deprive me of air, or common chemical foods, and my body would try to seek out other sources of energy—sunlight, radioactivity, or whatever is available. Even change my basic tissue fluid from water to—"

It sounded a little like bragging, so Jan cut in with a feminine tease: "Yes, Dr. Lanvin. But put on your overcoat. People will think it odd that you're carrying it on such a sharp winter afternoon."

Doc laughed back, and obliged her almost with embarrassment, and we were three old friends together.

"People get injured," I said, "or just grow old; and though limited rejuvenation and repair is possible, this is a far better way. That's how it should go, Doc; and you'd think that no one with sense would want to stop it. In months there'll be thousands of androids. But here we are again—unsure of how it'll all be taken. Like you say, this is succession to natural man. It can be conceived of as the old Threat of the Robot idea, with refinements. A force of staggering newness, wonderful to the point of being terrifying. We're almost certain that there'll be trouble."

The story of all we'd learned among the micro-Xians, and its repercussions here at home, was mostly regarded as a fantastic rumor at first. It was talked of lightly on the newscast, and wherever people gathered:

"Little People that have been around all the time, watching us? Shucks, even my Irish grandmother knew that! So we're gonna become wonderful, artificial critters! Homo ex Machina! Well, well!... Okay—take me—I was always one for improvements!"

Yes, it went something like that. And when people first truly knew, their reactions were mild, curious, and friendly. One incident I remember particularly.

Jan and I and Doc and a very pretty girl were walking in a quiet street near the University. The girl was someone I had known from a picture. She *looked* like the picture, again, now. That is, she had become like Doc. For the sake of youth and beauty, women can be more bold than men. She was Irma Tandray Lanvin, Doc's former wife—returned. And maybe she'd learned something about her man—that her rival, science, was part of him, and that she'd better take him as he was. Maybe he'd also learned the need of being attentive to a woman. Anyway, they both looked devoted, now, and I hoped it was so.

But what I meant to tell about was our neighbors. First we met Corbison, the mechanic, saying:

"Hi, Professor Lanvin. A fella'd hardly know you."

"It's still me," Doc answered.

Others gathered around as we paused to talk.

"How do you feel, Doctor?" asked an elderly woman. And when he replied, "Fine!" she said, "Think of it! I'm glad!"

There was even a pooch, who began with a prolonged sniffing at Doc, which progressed to a puzzled yelp, a wrinkling of brow above soulful and humorous brown eyes, then a licking of his hand, and a caper. In my mind the thought sprouted that a dog could become android, too.

"Wouldn't the word be 'canoid?'" Jan teased, knowing me well enough to be sometimes almost clairvoyant.

"Ah, the language struggles to keep up with progress!" a bookish youth commented lightly.

One of two small boys with their father fumbled with my fingers. "Aw, it feels just like anybody's hand, Mister," he growled, disappointed.

"That's a case of mistaken identity, young fella," I pointed out. "I *am* anybody—yet."

Irma Tandray Lanvin took his grubby mitt, and laughed. "Is that the same, Joey?" she questioned. "It shouldn't be, but I'll bet it is."

The kid looked as if his leg was being pulled.

There was just friendly interest and wonder among all those people, then.

"What they reminded me of," Irma said later, "was some kind of simple natives on a lost island, being shown a mirror for the first time—before they think of black magic. Is that what we all are, basically, at first? Simple? Trusting?"

"That's a good question," Jan commented.

And so it was for months more. But all the elements of catastrophe were present. Earth was a crowded but beautiful place. Technology had done much to give it an idyllic mood, and to shelter its inhabitants in cotton-wool. But that same technology that could build so miraculously, still held a devilish potential, if it served minds motivated by hate and fear. Need one even remember, here, the asteroids that were the fragments of Planet X, or the glassy, fused-down ruins of Mars, still slightly tainted with radiations of nuclear fusion and fission?

The drives of intellect, of whatever origin, seem always to have a sullen, combative streak, constructive in one sense, since it is the force that brings peoples up from nothing. But the stubborn taking of sides also harbors deadly danger.

Almost unobtrusively at first, the threatening clouds began to gather throughout the world. At our busy and expanding lab, Bowhart, who, with

Scharber, had been crewman aboard the *Intruder*, came to represent one phase of the opposition to the Great Change.

I remember what he said to me one day, his earnest face serious, his brow crinkled with the effort to be reasonable:

"Charlie, I could be all wrong. But for some time I've been thinking. Already there are twenty thousand once near-dead people who have been changed over; not to mention five thousand others who were in good health. Part of me admires the humanitarian angles here. But then there's that feeling of a slow, creeping invasion, so far unopposed. I can't exactly put my finger on just what makes it horrible; but at night I wake up sweating cold all over. Maybe I've got a blind spot in my head. All I know is that most everything about this remarkable duplication of humanity goes against the instincts in my slow Neanderthal guts. No, don't argue, Charlie. I've heard all of Dr. Lanvin's counter-points, and I just can't feel right about the whole thing. So I'd be a hypocrite if I worked in this lab any longer. I'll leave today, with the best of wishes to you and yours, and Dr. Lanvin. Tell him, will you?"

"All hail, Bow," I said, shaking his hand. "Thanks for the honesty. I know what you mean. I've felt it all myself, even though I don't quite agree."

Scharber, his former buddy, was also present in my office. They shook hands almost formally, now. For Scharber had moved all the way to the other side of the fence. He'd become the thrilled, eager kind.

"Poor Bow," he growled after Bowhart was gone. "A good guy, a gentleman. But mixed up, like some tough kid, afraid to ride on a merry-go-round. Feeling a black-rat-brown-rat difference. A primitive terror of being crowded out by something far more vigorous, and different from what he has always conceived of as human. Which brings up the reason why I'm here to see you, Charlie. I've screwed up my nerve to change the quality of my bones and meat. As far as I'm concerned, the process might as well start tonight. Okay?"

I nodded. "Okay. Fine, Scharber," I said.

If folks had all been like Scharber, there would have been no obstruction of progress. If they had all been like Bowhart, there would at least have been

no danger. But as always, there were other types. Among them were those who like to speak out against something.

Among these, now, was an old classmate of mine, whom I have mentioned before, one Armand Cope. Already he was becoming minorly famous, laying down the "facts" with a definite oratorical talent. I think that he was, in the main, honest in his beliefs. But pledged and prejudiced to one point of view, he was blindly violent toward its opposite number. Cope was a fanatic. And now, with the smokes of fear curling in many minds, nothing could have been more dangerous than his activities, and the activities of the numerous individuals who were like him.

I heard him speak over the radio and television. Always his words drummed on the same points:

"Friends, the craze for gadgets has become a folly, an insult to man's dignity. The proof has become brutally plain today. All we ever wanted was to live an uncomplex life—having houses that we build, and crops that we raise, with simple materials and simple work of our muscles, as nature intended. Science? Much of it should have stopped before it ever started. It was a trap from the first, offering its benefits as bait, not letting us know that it led to this mechanical abomination, which seeks to sully our own natural being with a hideous slime of the laboratory! The prospect makes one's nerves crawl; death is better than the triumph of such a thing! We must fight and fall, if necessary! Let the maniacs and fools know the real strength of humanity!"

Plenty of people were eager to listen to Cope, and to cheer him on.

I gulped, and then grinned at Doc rather wanly. Jan and I were in his house that particular evening.

"It's like we thought it would be, before anyone on Earth even knew about what we were bringing them," Jan said.

"You're going to talk back, Shane," Irma, Doc's wife, commented, with a thread of steel in her voice.

"Of course I'm going to talk back," he answered. "But I'm afraid that that could never do enough good. There'll always be enough point to what Cope and his kind say, for scared, furious souls to cling to. I wish mightily that it could be different; but I suspect that what I say will only help to consolidate another fierce belief, to oppose Cope's believers. Yes, like two mighty armies being drawn up for battle. That is the real danger! Well, anyway I've got to try."

And so Dr. Lanvin was on television the following evening, speaking from the Civic Center of Chicago. Jan and I left to run the lab, listened from my office. It was a good speech:

"... I've never liked cheap, showy gadgets, performing some small function that a person might do as well, and as easily, and with less affectation, with his own head and hands. There, perhaps Mr. Cope and I agree, as, no doubt we do about a pastoral simplicity when it is possible—the smells of rain and woods and gardens. But Cope forgets that, crowded as the Earth is, with its billions of mouths to feed, such beautiful, rustic inefficiency is no longer possible, and hence beyond being argued for, reasonably, unless the starship brings us to other habitable worlds.

"Which presents the subject of inventions—natural products of natural minds which are too sublime to be called gadgets. The starship, for one. The android process, for another. Does Mr. Cope suppose that the benefits the latter represents, would ever encourage mankind as a whole to suppress it? It couldn't be suppressed, by law or by anyone, as long as there are people left to dream of vigor going on and on.

"Mr. Cope says further that his nerves crawl. This is nothing more than the mistrust of the new and unknown, which time will take away. Yet, worst of all, he speaks of fighting and falling. I hope that he does not mean it. For today, that can truly be a thing of horror, and final silence. Therefore, I plead that he, and all those who have been tempted to think in this manner, review their reasoning, and correct its defects."

I visited Cope at his home. "Look, Cope," I said, "we used to be friendly enough to live and let each other live. Don't you see that what you're doing now can end all that has been built, and finish the human race—natural and android—entirely? You're bucking a logic and a need for betterment that's far too big for anyone—the death of death, you might say. What do you

want in its place? The death of everyone? You've got to stop talking as you do, Cope, pounding on the detonator of a world!"

His intellectual face went white with rage at what I had said. "You—Harver!" he growled softly. "You dare to talk to me like that! When you helped to turn this hellish development loose on Earth! Make every human being a snake, and it would not be half as bad. Yes, I was half your friend. But now get out of my house—out before I kill you!"

Further signs of danger were soon more definite, after that. Several days after Scharber's emergence from the process, I was walking with him in a Chicago street. A tactless acquaintance of his, of opposite inclinations and a dislike of him, previously entertained, ran into us in a theatre lobby.

"Hi, Scharber," he greeted. "I heard. You were born a robot, so why bother to change? And why didn't you at least order yourself a better face?"

Scharber retained a normal capacity for getting sore, and only a normal amount of self-control. "A robot is a machine, Powers," he said. "So is the old time protoplasmic man. So is the android. It's silly to make a distinction, based on silly pride at being what you seem to think of as exclusively human. And maybe your face could also benefit by some changes."

Sure, Powers had been brooding, too, and brewing up poison. The fact that he swung at my companion, proved it. Scharber ducked like lightning, and responded with a much-pulled return punch—if he'd given it half of full force, Powers' jaw would have been a mush of bone-splinters. Powers went flat; and it was some seconds before he started to scream and curse:

"Tin monsters!" he spat venomously and inaccurately. "Get them—both of them! Trying to crowd us off the Earth!"

Somebody with sense shouted, "Keep your heads!" But that, to some others, only represented the challenge of opposition. A half-dozen men came at us at once. I upset two of them all right; but being still just ordinary, I wouldn't have had much chance, if it wasn't for Scharber. Presently, with a pack gathering around us, we had to fight our way out of there, Scharber sprinting away at last, with me riding him pickaback. No protoplasmic man

could have run a third as fast as he did then. I suspect that that display of speed scared and infuriated our attackers, further.

Other androids came up against this same kind of experience, and their constant victories in such scuffles, sharpened their terrifying aspect in many minds, and the conviction that there had to be a battle to the death.

Nor was it only humans of the older order who gave way to outbursts of fury. Soon it was give and take. Androids retained all of the old capacities for various emotions. It seemed that each violent incident would be followed by something worse.

I saw one android blown to bits, his flesh still squirming hours after he had ceased to exist as a composite entity. One severed arm had drawn itself along the ground with clutching fingers, almost like a great slug crawling, for two hundred yards.

There was something demoniac about that, which, for the moment, almost made me agree with Armand Cope.

The fury of the conflict came to a head one night when our laboratory went up in a cloud of nuclear fire. Five hundred persons were wiped out in the blast. It was lucky, indeed, that the lab was outside of Chicago proper, or the casualty list would have been much longer. Of our inner circle of friends, only Scharber was in the blast, and he escaped flying fragments and incandescent heat by dropping behind some heavy masonry. Radiations couldn't hurt him at all, though for a time he must keep away from the rest of us. The others of our group were safe in town.

There was the cold rage in Scharber's face when I first spoke to him from a little distance at the edge of the ruins.

"Damn them all, Charlie!" he growled. "Stupid, thick-headed, backward fools!"

"Easy, Scharb," I said. "The government, and the considerable majority of saner people, are trying to restore order."

It was true. Police forces were everywhere. Our president pleaded for calm. A cache of nuclear munitions was discovered and put under guard. It might even have belonged to androids. Nobody knew. It was in an old Chicago cellar. But of one thing we were sure—that there had to be many other

caches of hellstuff, undiscovered and available to the hotheads and jerks, hidden in caves and woods and various other places, throughout the world.

One thing wasn't done. Armand Cope, and other rabble-rousers like him, were not put under restraint. It could have been accomplished within the emergency provisions of democracy, though a willful connection between the speeches that they had made and the blowup of the lab, could not be proven. Maybe the government was afraid to restrain them—afraid that their arrest would make them martyrs—and that this martyrdom would trigger the bombshell in the taut nerves and frightened minds of their followers. This belief may well have been the truth.

IX

Jan and I went to Doc's house, inside a police cordon, for a discussion. We risked radiation by bringing Scharber along. We wanted to make sure that he wouldn't do anything vindictive, which might well have happened had we left him by himself.

Irma met us at the door. "Shane almost wishes now that the android process had remained just the property of the micro-Xians," she said. "That's how bad matters seem to him at this point."

Doc jumped to his feet as we entered his study. "Cope means to speak again tonight," he announced. "Cope, and about a hundred others of his crowd, from scattered radio and television stations. We know about what they'll say, more or less. Yeah—'Get rid of these mechanical demons while there are still less than thirty-thousand of them. Before it's too late! Kill the serpent! Return to simplicity! Do you know that even their radioactive metabolism is poisonous to us?'"

Doc paused and groaned. "The latter isn't even true," he went on. "At least not while an android is on Earth, breathing oxygen and living by chemical energy. Then the radiation of a subatomic tissue-process is suppressed almost to zero. But that's the way most of Cope's arguments go—they leap thinly to conclusions, without thinking matters out to any depth. But many people don't want to think deeply, or else they're too frightened. And tonight

I suspect that Cope and his bunch will give the order to attack. Charlie, what are we going to do?"

I was in a cold sweat. "You know what we can try as a temporary relief measure, Doc," I said. "We can silence Cope and a few of the others—you know how. The only trouble is that there are so many of those loudmouths, and only you and I and maybe Jan who are in a position to do the only thing that can be done. We may not be able to shut up anywhere near enough of them to get over this danger spot, but we have to try."

Jan came over to me and pressed my hand, and it helped. She was always courageous and cool.

As it turned out, there were few speeches of Cope's kind made that night. Cope collapsed before the television lenses and the microphones. No, he didn't die; he had what looked very like an epileptic fit. He dropped before he uttered a word. He frothed at the mouth, he snored. He looked ridiculous, even mad.

Why all this happened was simple. It was an old Xian trick. A micro-android—Doc had transmigrated briefly again—was inside Cope's skull, tampering with his brain. The tiniest flash at lowest power from a jet rod directed against the proper nerve center, was how it was done.

Doc silenced another character called Minton. I gagged another pair of flannel-mouths named Trefford and Donalds the same way. Jan managed to fix one called Parkhurst. That made five of the worst who had been operating around Chicago. But it still left over ninety others. It worried us badly, until we got back home, and into normal-sized bodies, once more. Scharber had been a good boy, staying out of trouble beside Doc's television, with Irma.

"Not one of the others said much either," he announced quietly. "They all fell on their faces the same way." He paused for just a second before he added, "I wonder why?" his eyes oddly aglow.

"There could be only one answer to that, couldn't there?" Irma hinted.

Doc grinned reminiscently.

Jan smiled. "The elves of legend, the helpful ones," she chuckled. "Well, who knows but what there's a connection with those old folk tales? Legends

frequently have a basis in fact. It seems that I remember a strange, deep little guy who lives way out in space, and down near the limit of smallness. His name was Kobolah, and lots of his people didn't believe that Earthians should be trusted. He almost got into trouble over that. But it appears that he still has lots of friends among his own kind who'd like to see the android become successful among us. It seems, further, that if Kobolah's particular asteroid world took off for the stars, already, as appeared to be intended, he and some pals have so far stayed behind. Or else it was just some pals of his who helped us. But who knows? Maybe we'll see him again. Anyway, his world was as wonderful a place as you could imagine. I wonder if there's anything more strange in the whole universe?"

As Jan's musing words ended, I saw a strange, speculative look in Scharber's face. Doc's eyes were soft for a second.

"I guess that miniature things still intrigue me," he said. "But we're tied up with bigger facts now. I think we've won a temporary peace, but I'll bet that that's all it will be—temporary. Even if Cope and the rest of the same crop stop shouting, now, there'll be others to do just as they did. In a day or two we'll know for sure."

Doc was right. On the very next evening Armand Cope was on the air again, frightened, but determined. "This treachery of last night, even though I do not understand its method, makes me even better aware that this is a fight to the finish," he growled. "A fight against a hideous thing, to which there can be no end except victory or death. As long as I am a man, I shall be proud...."

Doc shrugged mildly. "I'd almost say 'Blah, tiresome fool!'" he remarked. "But it wouldn't be fair. Cope stubbornly believes what he says, I'm sure. It's etched into his nature. To a lesser degree with most, it's the same with many others. So, this is it."

The following evening, Doc made his suggestions over the air, speaking from his house:

"I am addressing those, who, in the eyes of some, have ceased to be human. But perhaps the term, 'android' should be dropped entirely. We are men in form, mind, emotion, aim, and pleasure—let there be no instinctive, sullen, backward doubt of that! Our shape and our organs are human. We have sprung from man's aspirations, and his quest for more knowledge and better living. Though the knowhow of our living was borrowed from another people, it would have come to men on Earth in time, and by their own efforts. We are thus, simply, a far hardier variety of what humans have always been. To those who are weaker, troubled by fear, less understanding, we should be generous, until more time lets them realize these truths. Therefore, I suggest that we leave the Earth to them, going outward where our powers permit us to go freely."

That is how it has been. Among the androids, as if the interstellar regions was their natural habitat, Dr. Lanvin's hint took hold at once. On Earth, tension eased gradually, until even Armand Cope's voice sounded puzzled, and then sank to silence.

But let me tell about a side-event. Doc found a toy-sized craft in his workshop, a ship with tapered bow and stern, and retractable airfoils. It was less than an inch long. Need I say how we boarded it—Doc, Jan, and I? Or how later, we and one Kobolah, conversed under the scope of a micro-manipulator, while Scharber and Doc's Irma took turns watching us through the lenses?

We thanked the tiny Xian for all his help. We saw his electronic visual filaments blink over his eyes when Jan suggested:

"Kobolah, you could be cast in a larger form like the old Xians. You could go with Dr. Lanvin in the first ship to leave for the solar system of Sirius."

"Maybe—someday," he buzzed in answer. "Not now. To Sirius? I'm going there, anyway with my own people soon. Time? There is plenty—for everything. May you make few errors."

Then, with his jet rod he blasted off into the air. Within a minute, his ship, aboard which were hundreds of his kind that we had seen, spat blue fire, and darted out of the open window.

Scharber chuckled almost wistfully. "Micro-androids," he said. "Strangest thing I ever saw. Why didn't he take me with him? Got to start seeing the

Dear Dr. Lanvin, Mrs. Lanvin, Charlie, Jan. Everybody—So I win.... The Little Guy must have guessed. Anyway, he brought his ship here. Then he wrote his questions—though he could hear me answer. Do I want to come along? Yeah—look at the other papers—see for yourselves. You must have made a good impression out there—you who were there. So he likes Earthlings. For pets, maybe? Who knows? Well—I didn't say no.... Wish me luck, and the same to you. Do me a favor? Whoever goes first out to Sirius, take this big carcass of mine along—being android, it ought to keep for a long time. Maybe I'll need it after a while. Right now I'm getting a smaller edition. So long for maybe a hundred years, more or less.

Scharber.

Smiling like an elf, Doc looked at me. "How do you feel?" he asked.

"Same as you, I suppose," I answered. "Haunted...."

During the year that followed, that first starship was completed, and ten others of the huge mile-long craft were begun. Jan and I saw them all in their cradles when we went out to the Moon to visit my mother and dad.

It was really meant to be a farewell trip. Jan and I hadn't expected to get berthed on that first starcraft, the *Euclid*, but it happened. Not all of the voyagers were of the new flesh.

"Farewell nothing," Dad told me slyly at the house. He looked more like a slightly older brother of mine, than somebody paternal.

"We're going along, Charlie," Mom intimated. "We've always been ready for new adventure, haven't we?"

In due course the *Euclid* came to the New Mexico Spaceport to pick, up its passengers, Jan and I and the folks had been on Earth for over a month by then. We and Doc and Irma arrived at the port on the same rocket plane, and as I looked up at the brooding hull of that colossus I felt a little as if a kid dream of mine had come true—that I was matching my lusty strength against the whole universe, and winning. To fight and to win against something, has been a need in human blood and bone for uncounted eons.

But should I feel a bit puny and sheepish, too? Comparing myself to Doc, for instance?

This was his special day. Back there behind us, as we approached the starship—back there beyond the guardropes—were the crowds of curious, thrilled, scared, envious humanity. Some cheered for what the *Euclid* meant to progress—or perhaps they cheered more for a greater triumph—the thirty-thousand demigods who would be among its passengers. But was some of the cheering given in relief at being rid of them?

Bowhart was there, to shake hands with Doc and me and Irma and Jan, and to meet my folks.

"Good luck to you all," he said. "No Great Change, yet, for you, Charlie? So I hear. Funny, hunh? Dr. Lanvin—I want to give you special best wishes. You look happy, so I guess if you're satisfied, nothing I can say will be an offense. But I still wouldn't want to be you for a million dollars."

Bowhart must have known that much, saying what he did; because Doc wasn't at all offended—just airily nettled, like an ageless leprechaun pitied by an urchin.

"Oh?" he asked lightly. "In the past many a millionaire would have given more than a million for another week of life and vigor, and it was no sale. The value is a lot bigger; but it doesn't cost that, now—it doesn't cost anything except a little more growing up. What do you want to do, Bow? Drink beer, eat ice cream, make love? I can do all that, too. Someday you'll get it through your fuddled head that I'm still human. I think you're catching on already. Yes, the androids are leaving Earth; but you know that the process that makes them is still here. Every day there are more labs. Because people get hurt terribly, or wear out beyond reasonable repair. And what would you expect them to want to do then, just die?"

Doc wasn't just talking to slow minded Bowhart, but to all humanity that was like him. It was his final message. But there was another touch to it that wasn't in words. It was a cocky gentle air that maybe suggested the contrast of—say—eating a fine dinner, and then taking a long dive, unclothed, through the vacuum of space—both with equal relish.

Bowhart looked puzzled, and a bit sullen. Maybe he was beginning to catch on at last.

Well, we made that enormous jump across the light-years to the Sirian System. Seventy-nine years it took. I don't think that even an Xian ship could have done much better. There's no overdrive or time-travel in sight. Funny, isn't it—here, for once, nature resists us. But to avoid boredom there was the older idea of suspended animation—natural to the android, and capable of being induced in the older flesh by special anesthetics and chilling. My wife and our friends passed the first two years of the journey awake, to help operate the ship. The other seventy-seven years passed as a moment.

We found us a world just slightly smaller than the Earth, and young and beautiful. There was no native intelligence yet, comparable with the human. The valley in which we live is rich and lush, and it slopes down to the ocean. Like my dad and mother, Jan and I have a sturdy house of stone; cleared fields, and livestock descended from the animals and poultry brought out from Earth.

It's Mom's old rustic dream. It's even Cope's! It's an idyll.

A town is springing up fast nearby. It is one of the first colonial settlements of what may become a great Earthborn interstellar union.

Doc is in the town with Irma, building it, planning, full of goodwill for everyone. Scharber's normal-sized android body still sleeps in a special vault under the town hall. But who knows at what moment he and Kobolah may come?

Doc kids my folks and Jan; but especially he kids me:

"You're silly, Charlie, why don't you switch over to the android level? What are you waiting for? Sure, I like to live in a house, too; but sometimes I sleep out in the rain or the snow just for the hell of it! Of course there's no real good in that kind of nonsense! But changed over, a man has an average of a twenty per cent increase in intelligence, simply on the basis of better energy and alertness! You may think that you feel good, but even if no trials come to demand superior stamina, you'll feel better; you'll do three times the work, and never tire at all! Why, even on Earth, according to reports that are relayed from starship to starship coming this way in a long string, humans as they were are almost gone. So what are you—a diehard, a stick-in-the-mud? Even—you?"

"Maybe it's the seventy-seven years lost, Doc," I josh back at him. "I've got to catch up, perhaps. Of course I recognize all the advantages. I've been through the mill. Just as with you, in my head, lodged against my upper skull and doing me no harm, the medics say, is a micro-android which my ego has inhabited, and which I almost never use now. I remember what it is like to be super, Doc. I grant that it is all the truth. But there's time. Just let me think some more."

Yeah, Jan and I think of all we've seen, that we never dreamed was there. Beauty, strangeness, vastness, smallness, wonder, knowledge. We've come a long awesome way.

You feel that you know a little more about the universe, and that you're warmly and humbly a little nearer to its ultimate Mystery, and are at peace. You know that the Great Change in man is right, and was intended.

We've been stubborn, and I'm not entirely sure why. I know that we and the others of the old flesh will yield to progress sooner or later. Maybe we've been clinging sentimentally to the past of man. But deep down, I believe I know the real reason. We're slow, we're human; just give us time. It's hard to accept the responsible role of demigod.

We're just scared of so much newness.

*** END OF THE PROJECT GUTENBERG EBOOK DAWN OF THE
DEMIGODS ***

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